# Dammed in the Mekong: Averting an Environmental Catastrophe

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# Principal Findings

**What is happening?** Exploitation of the Mekong River basin's natural resources, especially through hydropower and sand mining, is causing escalating harm to the region's ecosystems and endangering the livelihoods of millions of people. The effects of development are distributed unevenly, with many of the costs borne by the poor and marginalised.

**Why does it matter?** Environmental damage from rampant hydropower development and climate change threaten the river's seasonal movements, supply of sediment, fish stocks and the physical security of people living along the Mekong in Cambodia, Laos, Thailand and Vietnam. These populations could be at risk of large-scale displacement.

**What should be done?** Policymakers should pay greater heed to the ecological value of the natural flow of the Mekong and its tributaries. They should accelerate adoption of solar and wind power. Greater transparency, data sharing and participatory decision-making – including consultation with riparian communities – in infrastructure development would improve transboundary resource governance.

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# *Executive Summary*

Environmental degradation caused by infrastructure development in the Mekong basin, especially large-scale hydropower and sand mining, threatens the region's ecological balance and the livelihoods of 70 million people. Over the past 30 years, developers have largely ignored the natural ecosystems' environmental and social value, while the emergence of new infrastructure and industries has deepened inequality within states. None of the Mekong basin states – China, Myanmar, Laos, Thailand, Cambodia and Vietnam – has achieved a satisfactory balance between the imperatives of economic growth and environmental preservation, while climate change has resulted in higher temperatures and severe droughts, exacerbating the harmful effects of new infrastructure. Greater understanding of environmental change is needed, with large-scale hydropower giving way to solar and wind. More effort should be invested in transboundary governance of the basin's resources and in strengthening the participation in decision-making of the communities hardest hit by damage to the Mekong.

China's cascade of twelve mainstream dams in the upper basin are arguably the source of the greatest environmental harm. But the countries of the lower Mekong basin – Laos, Thailand, Cambodia and Vietnam – are hardly blameless when it comes to damming rivers. Laos has staked its economic development on hydropower. There are already 75 dams on Lao tributaries, with 61 more planned, and two dams on the Mekong mainstream, with at least seven more on the drawing board. Thailand is the largest investor in Lao dams and the largest purchaser of Lao electricity. Cambodia announced a moratorium on Mekong mainstream dams in 2020, but it continues to dam tributaries. Vietnam, which bears the brunt of downstream impact of this construction, is itself an upstream dam builder, and Vietnamese public-sector companies invest in Lao dams. Sand mining, particularly in Vietnam and Cambodia, also threatens the delta, causing riverbank and coastal erosion and exacerbating salt-water intrusion.

People living along the Mekong and many of its major tributaries are already contending with irregular fluctuations in water levels, declining fish catches and the loss of traditional livelihoods. Studies and modelling point to the dire and intensifying environmental harms of hydropower and other human activity on the basin. Dams trap sediment that is crucial for nourishing soil, flora and fauna, with knock-on effects on the basin's ecosystems and biodiversity and particularly on rice farming. Fragmentation of the river also disrupts fish migration and reproductive cycles, threatening the world's largest freshwater fishery and endangering a vital source of protein for local communities.

Irregular flows of water and sediment pose a risk to the Mekong's annual flood pulse, which normally sees Cambodia's Tonle Sap Lake expand six-fold in size. This annual phenomenon is vital to sustaining Mekong fisheries, healthy wetlands and agricultural productivity. Reduced flows of water (in the wet season) and sediment also compound land subsidence and salt-water intrusion into the Mekong Delta, home

to seventeen million people and Vietnam's most productive agricultural region, most of which may be submerged by 2100.

The social and political consequences of projected environmental crises are necessarily speculative, but the worst-case scenarios are alarming. Over time, destruction of livelihoods could spur large-scale involuntary migration, boosting illicit economies, undermining government legitimacy and fomenting social discord in large parts of the lower Mekong basin.

Oversight, regulation and transnational cooperation to address the threats to the Mekong's future are sorely missing. To the credit of the basin's states, tensions between upstream and downstream states over water use and other conflicting interests simmer but have so far not escalated, despite the intrusion into the Mekong of major-power competition between China and the U.S. That said, cooperation among states has not always improved matters, instead often compounding environmental damage and smothering public participation in decision-making.

Shunting the costs of development onto the poor and marginalised, frequently ethnic minorities in their respective countries, has become standard practice across the Mekong. These communities, whose livelihoods and identities are bound up with the river and its tributaries, are most directly affected by environmental changes, yet largely lack the capacity to influence policy. Decision-making in the region is top-down and participatory mechanisms are weak. Civil society groups have challenged development policies and advocated for alternatives when they can, notably in Thailand, but the region's governments have accorded them little space and given them a limited hearing.

The Mekong River Commission, the basin's only treaty-based organisation, has no regulatory role, as its member states are unwilling to cede sovereign authority to supranational bodies. Meanwhile, states and institutions across the region have adopted the rhetoric of "sustainable development" without honouring its substance. Hydropower is being driven by governments and international networks of private enterprises and investors who are largely divorced from accountability mechanisms and local knowledge of riverine livelihoods. These actors have deep vested interests in expanding hydropower regardless of the harm caused.

The failure of governments thus far to grapple with the looming environmental catastrophe is not for lack of scientific consensus. There is broad agreement among researchers on the damage caused by large-scale hydropower as well as measures that could improve development planning and help safeguard the environment. These include ensuring participatory governance to involve local communities in decision-making; conducting research and sharing data at a basin-wide level; and accelerating adoption of solar and wind power, which are increasingly affordable and available at scale, to substitute for new hydropower. More broadly, all new development plans along the river should be informed by an appreciation of the value of the basin's natural ecosystem and the long-term costs of harnessing nature for short-term economic gains.

A more fundamental problem is that sensible recommendations to protect the environment are meaningless without policy processes that could accommodate them. Comprehensive solutions targeting the causes of the basin's degradation must contend not only with other priorities backed by national governments, but also con-

straints on the ability of states to coordinate across public and private sectors and national boundaries. Policymakers would better serve the public – not just those living alongside the Mekong but the many millions who depend on its resources across mainland South East Asia – by anticipating the problems of over-exploiting the river and its tributaries than by reacting to the catastrophic consequences.

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## I. Introduction

The Mekong River, the longest in South East Asia, rises in the Tibetan Plateau and flows roughly 4,800km through the upper basin of China and Myanmar and the lower basin of Laos, Thailand, Cambodia and Vietnam.[1] It is the world's largest freshwater fishery, with an annual wild-capture catch of some two million tonnes, or 17 per cent of the global inland fish harvest, valued at $11 billion.[2] Home to at least 600 freshwater fish species, with many more to be discovered, the Mekong is the third most biodiverse river system in the world, behind the Amazon and the Congo.[3] Of the estimated 70 million inhabitants of the lower Mekong basin, 75 per cent depend on fishing and farming for their livelihoods.[4]

Though it is the lifeblood of populations in six nations, the Mekong faces a convergence of threats, notably from environmental degradation and climate change, with prospects for tackling them complicated by national politics that gives little voice to riparian inhabitants as well as by geopolitical rivalries. On today's trajectory, resource scarcity seems a probable scenario, potentially driving political instability.[5] Economic growth has brought with it growing dependence on hydropower from the basin's rivers as well as other sorts of resource exploitation, generating accelerating environmental costs. Despite declining rates of poverty and high rates of GDP growth over recent decades, the poor and marginalised communities living along the region's rivers have disproportionately borne the costs of environmental degradation.

Competition between the People's Republic of China and the United States has also highlighted the tensions between upstream China, which has built twelve hydropower dams on its section of the river's mainstream, and the downstream countries of the

[1] The Mekong basin is roughly 795,000 sq km.

[2] Wild capture refers to the harvest of wild fish as opposed to farmed fish. Ian Campbell and Chris Barlow, "Hydropower Development and the Loss of Fisheries in the Mekong River Basin", *Frontiers in Environmental Science*, vol. 8 (2020), p. 2.

[3] Ian G. Baird and Zeb S. Hogan, "Hydropower Dam Development and Fish Biodiversity in the Mekong River Basin: A Review", *Water*, vol. 15, no. 1352 (2023), pp. 2-3.

[4] One third of the basin's population lives in the Cambodian floodplain and the delta that spans the Cambodia-Vietnam border. Cambodia and Laos are among the poorest nations in South East Asia, with per capita GDP estimated at $2,630 and $1,980, respectively, in 2024. See "GDP per capita, current prices", International Monetary Fund. Of the basin's 70 million total inhabitants, some 29 million live within a 15km corridor on either side of the Mekong, including 2.1 million living within a 5km range. "MRC Strategic Environmental Assessment of Hydropower on the Mekong Mainstream: Summary of the Final Report", International Centre for Environmental Management, 2010, p. 16.

[5] Crisis Group Asia Report N°332, *Transnational Crime and Geopolitical Contestation along the Mekong*, 18 August 2023. See also Suwit Laohasiriwong and Mikio Oishi, "Managing the Mekong River Conflicts: Political Stability at the Cost of Local Communities", in Mikio Oishi (ed.), *Managing Conflicts in a Globalizing ASEAN* (Singapore, 2017), p. 144.

lower Mekong basin – Cambodia, Laos, Thailand and Vietnam. Some have suggested that the region is analogous to the South China Sea, an arena for intensifying geopolitical friction.[6] But there is much more to the tensions in the Mekong basin than geopolitical competition, which does little to explain the conflicting interests among the downstream states, as well as disputes between communities directly affected by infrastructure development, on one hand, and states and business interests, on the other.

This report unpacks the issues and forces at play in the Mekong to gain a clearer understanding of the competing interests in the region and associated risks to human security. It is based on review of existing literature, interviews held between June 2022 and January 2024 with experts, academics, former officials and people directly affected by the basin's changing environment, including women, in Cambodia, Laos, Thailand and Vietnam. The report does not cover Myanmar, which is situated almost entirely outside the lower Mekong basin.

[6] Crisis Group interview, Western diplomat, Hanoi, Vietnam, December 2022. See also Catherine Wong, "Is Mekong River set to become the new South China Sea for regional disputes?", *South China Morning Post*, 3 January 2018; David Hutt, "Mekong River: A new flashpoint in Indo-Pacific?", Deutsche Welle, 12 August 2021.

## II. Commodifying the Mekong

As a rule, the Mekong states tend to regard natural resources as inputs to be transformed into economic growth, with little concern for the environment or the equitable distribution of benefits.[7] Hydropower and other development activities have turned the river basin's natural resources into commodities and channelled the resulting revenues and other returns to urban elites.[8] What was a free-flowing river system in the early 1990s is increasingly engineered and fragmented.[9]

### A. *Hydropower*

Hydropower has been attractive to Mekong states as a way to propel economic development, meet rising electricity demand generated by population growth and industrialisation, attract foreign direct investment, expand irrigation, control flooding, reduce drought, generate revenue and, ostensibly, reduce poverty. Proponents justify hydropower as a green and sustainable energy source, with storage capacity necessary to complement intermittent renewable sources. There are at least 661 operational dams in the Mekong basin, of which 447 are small non-hydropower dams for irrigation, water supply and flood control. Operational dams have a total capacity of 36,058 megawatts (MW), up from just 1,242 in the 1980s. But with a further 54 dams under construction and 331 more planned, this basin-wide capacity is expected to double in the 2030s, with Laos accounting for the highest growth (18,223 MW).[10]

China controls not only the source of the Lancang River, as the Mekong is known in Chinese, but also the portion of the basin best suited to hydropower generation. It began developing a cascade of twelve major dams on the Lancang in the mid-1980s, as part of a campaign to develop its poorer western provinces and supply electricity to its booming, industrial eastern coast. At least two dams, Xiaowan and Nuozhadu, have reservoirs with multi-season regulating capacity, storing enough water to ensure power generation during the dry season, when demand peaks due to increased use of air conditioning.[11] Six more dams are planned on the river's upper reaches.[12] China's

[7] Scott W.D. Pearse-Smith, "The Return of Large Dams to the Development Agenda: A Post-Development Critique", *Consilience: The Journal of Sustainable Development*, vol. 11, no. 1 (2014), p. 125.
[8] Karen Bakker, "The Politics of Hydropower: Developing the Mekong", *Political Geography*, vol. 18, no. 2 (1999), p. 212.
[9] Crisis Group interview, environment expert, Chiang Mai, Thailand, January 2024.
[10] By way of comparison, the Amazon basin (7 million sq km) has 158 hydropower dams, with another 351 proposed. "Reducing negative impacts of Amazon hydropower expansion on people and nature", press release, Cary Institute, 17 February 2022. Wei Jing Ang et al., "Dams in the Mekong: A Comprehensive Database, Spatiotemporal Distribution and Hydropower Potentials", *Earth System Science Data*, no. 16 (2024), pp. 1217, 1223.
[11] Darrin Magee, "The Dragon Upstream: China's Role in Lancang-Mekong Development", in Stina Hansson et al. (eds.), *Politics and Development in a Transboundary Watershed: The Case of the Lower Mekong Basin* (Dordrecht, 2012), p. 176. China's Mekong dams have a total reservoir volume of 45 billion cubic metres and hydropower capacity of 21,250 MW. Should all planned dams be completed, this will increase to 63 billion cubic metres and 30,566 MW. Ang et al., op. cit., p. 1217.
[12] These are Linchang (72 MW), Yuelong (129 MW), Kagong (240MW), Rumei (2100 MW), Guxue (1,700 MW) and Gushui (2,600 MW).

Lancang dams paved the way for mainstream dams in the lower basin, not only through the example they set, but also because the altered hydrology led to more predictable water flows, improving generating capacity and thus the economic viability of downstream dams.[13]

Laos, accounting for 40 per cent of the basin's hydropower potential, is the locus of dam building in the lower Mekong.[14] Poor and landlocked, with a population of only 7.5 million, the country has staked its development on hydropower to generate foreign exchange and contribute to poverty alleviation. It has 77 operational hydropower dams, two of which – Xayaburi and Don Sahong – are on the Mekong mainstream.[15] Of 61 planned dams, seven are on the mainstream, all intended to generate electricity primarily for export and funded by foreign investors.[16]

In the 1980s, environmental activism in Thailand, most famously protesting the Pak Mun Dam in Ubon Ratchathani province, began to impose constraints on resource extraction and hydropower, inducing Thai companies to seek projects in neighbouring countries, particularly Laos, instead.[17] The Thai private sector is the largest investor in Lao hydropower, and Thailand's electricity authority has concluded Power Purchase Agreements for both Lao mainstream dams and several prospective mainstream dams in anticipation of future needs, despite enjoying an ample electricity surplus.[18]

Four of Cambodia's five largest dams are on tributaries to the Tonle Sap Lake in the Cardamom Mountains, in the country's south west. The 400 MW Lower Sesan 2 Dam, on a major Mekong tributary, is the country's largest, built by China's Hydrolancang International Energy and commissioned in 2017.[19] In March 2020, citing environmental concerns, Cambodia announced a ten-year moratorium on mainstream dams, shelving

[13] Darrin Magee, "Powershed Politics: Yunnan Hydropower under Great Western Development", *The China Quarterly* (March 2006), p. 30; Philip Hirsch, "Cascade effect", China Dialogue, 8 February 2011.

[14] Ang et al., "Dams in the Mekong", op. cit., p. 1220.

[15] Ming Li Yong, "Energy importers must consider true 'sustainability' of Laos hydropower", The Third Pole, 23 August 2022.

[16] These dams are, from north to south, Pak Beng, Luang Prabang, Pak Lay, Sanakham, Pak Chom, Ban Koum and Phou Ngoy. Ome Chattranond, "Battery of Asia? The rise of regulatory regionalism and transboundary hydropower development in Laos", Ph.D. dissertation, Erasmus University Rotterdam, 2018, p. 152; Ang et al., "Dams in the Mekong", op. cit., p. 1217.

[17] Natalia Scurrah and Philip Hirsch, "Land and Capital across Borders: A Regional Geopolitics", in Philip Hirsch et al. (eds.), *Turning Land into Capital: Development and Dispossession in the Mekong Region* (Seattle, 2022), p. 14. The Pak Mun dam was so damaging to local fisheries that it served as a case study in the 2001 World Commission on Dams report that spurred a global backlash against hydropower.

[18] Vincent Merme, Rhodante Ahlers and Joyeeta Gupta, "Private Equity, Public Affair: Hydropower Financing in the Mekong Basin", *Global Environmental Change*, no. 24 (2014), p. 22; "Thailand and neighbours sign pact on Mekong development amid dam fears", *The Nation*, 7 April 2023.

[19] Hydrolancang holds a 51 per cent stake, Cambodia's Royal Group 39 per cent and Vietnam's EVN International 10 per cent. A build-operate-transfer project, the dam will be transferred to Cambodia after 40 years. Oliver Hensengerth, "Regionalism, Identity and Hydropower Dams: The Chinese-Built Lower Sesan 2 Dam in Cambodia", *Journal of Current Chinese Affairs*, vol. 46, no. 3 (2017), p. 96; Mira Käkönen, "Fixing the Fluid: Making Resources and Ordering Hydrosocial Relations in the Mekong", Ph.D. dissertation, Faculty of Social Sciences of the University of Helsinki, 2020, p. 15.

plans for two large and potentially damaging projects, the Stung Treng and Sambor dams.[20] The government, however, continues to build dams on tributary rivers.

Located farthest downstream, Vietnam feels the impact of this accumulation of Mekong dams most acutely, but it is also an upstream hydropower operator. It runs at least 81 hydropower dams on tributaries or main branches of the Sesan and Srepok Rivers that flow westward from the Central Highlands through Cambodia and into the Mekong at Stung Treng.[21] Along with the Sekong, these rivers form what is called the 3S basin, contributing about 20 per cent of the water and 15-40 per cent of sediment to the Mekong River.[22] Vietnam is further implicated in upstream dams as it imports electricity from China and Laos, and its state-owned banks and businesses finance and build Lao hydropower projects, including the Luang Prabang dam under construction on the Mekong mainstream.[23]

### B. *Sand Mining*

In rapidly industrialising and urbanising South East Asia, demand for sand for use in construction and as landfill is high and increasing. Sand is mined throughout the lower Mekong basin but is most intensively extracted in Cambodia and Vietnam. A 2022 study of sand mining in Vietnam's Mekong Delta estimated that an average of 43 million cubic metres was extracted annually between 2015 and 2020, an amount some 53 to 146 per cent greater than earlier estimates.[24] Rates of extraction are fourteen to seventeen times the rate of replenishment and, at the current pace, will exhaust the stocks of exploitable sand in the Mekong within ten years.[25] The depletion is particularly alarming for the region's agriculture, as sand mining decreases riverbed elevation, and therefore water levels, among other effects.[26]

[20] Prak Chan Thul, "Cambodia halts mainstream Mekong River dam plans for 10 years, official says", Reuters, 18 March 2020. In November 2023, the new prime minister, Hun Manet, reiterated Cambodia's policy not to build dams on the Mekong. "Cambodian PM says won't develop dams on Mekong River", Agence France-Presse, 30 November 2023.

[21] "Sustainable Hydropower Development Strategy", Mekong River Commission, 2022, p. 17; Gerard Sasges and Alan D. Ziegler, "We Have Eaten the Rivers: The Past, Present and Unsustainable Future of Hydroelectricity in Vietnam", *Sustainability*, vol. 15 (2023), pp. 6-7.

[22] "3S Basins take another step towards shared long term trans-boundary vision", Mekong River Commission, no date.

[23] State-owned PetroVietnam Power is one of the partners in project along with PT (Laos), CH. Karnchang (Thailand) and CK Power (Thailand). "Laos works to put Luang Prabang hydropower project into use in 2030", Vietnam News Agency, 21 February 2023.

[24] Charles-Robin Gruel et al., "New Systematically Measured Sand Mining Budget for the Mekong Delta Reveals Rising Trends and Significant Volume Underestimations", *International Journal of Applied Earth Observations and Geoinformation*, vol. 108 (2022), p. 10.

[25] Lan Mercado and Marc Goichot, "Time to change myopic view of the Mekong", *Asia Times*, 4 April 2023; Le Dinh Tuyen, "Viet Nam's Mekong Delta is running out of sand. What's next?", Mekong Eye, 6 November 2023.

[26] Marc Goichot et al., "Sand mining and climate adaptation", in M.-N. Woillez and E. Espagne (eds.), *The Mekong Delta Emergency, Climate and Environmental Adaptation Strategies to 2050* (Paris, 2022), pp. 60, 65.

Cambodia and Vietnam banned exports of Mekong River sand in 2009 and 2017, respectively, but regulation of sand mining in both countries is uneven at best.[27] In Vietnam, sand miners are licensed by provincial natural resources agencies, but illegal mining is rife.[28] "There's a mismatch between government words and what the 'big guys' get away with", noted an environmental researcher in the delta.[29]

[27] Crisis Group interview, former member, Vietnam National Mekong Committee, Hanoi, Vietnam, 9 December 2022. Beth Timmins, "How the scramble for sand is destroying the Mekong", BBC News, 19 December 2019. Most of Cambodia's sand exports went to Singapore, where land reclamation has increased the city-state's area by 20 per cent compared to 1965. Between 2007 and 2016, Cambodia reported exporting nearly three million metric tonnes of sand to Singapore, but Singapore reported importing over 80 million metric tonnes from Cambodia. "This gap hints at the leaky, opaque nature of sand supply chains, where a few actors may make huge profits". Melissa Marschke and Jean-François Rousseau, "Sand Ecologies, Livelihoods and Governance in Asia: A Systematic Scoping Review", *Resources Policy*, vol. 77 (2022), p. 6.

[28] An Binh and Ngoc Tai, "Illegal sand mining crackdown cuts construction material supply", *Vietnam Express*, 3 September 2023.

[29] Crisis Group interview, environmental researcher, Ho Chi Minh City, Vietnam, August 2022.

## III. Environmental and Social Harm

The lower Mekong is characterised by an annual flood pulse caused by the change in water levels (up to 10m) of the Mekong in the wet and dry seasons. During the wet season, the increasing water level and volume causes the Tonle Sap River to reverse direction, usually in September, flowing upstream to fill Cambodia's Great Lake, or Tonle Sap. While the flood's intensity can vary, it generally causes the lake to increase six-fold in surface area, from 2,500 to 15,000 sq km. The Tonle Sap at full flood sustains the region's rich flora and fauna, including the fish that make up a large share of Cambodians' diet, and distributes the nutrient-rich sediment that replenishes the soil of floodplain and delta rice paddies. The inundation of Cambodia's plains also moderates flooding in the delta in southern Vietnam, and supplies water for irrigation during the dry season. In the dry season, the flow of receding flood waters into the Mekong Delta reduces salt-water intrusion and recharges aquifers.[30]

### A. *Fisheries*

Tens of millions depend to some degree on Mekong fisheries for food and livelihoods. But the Mekong's fisheries are under threat from dams, irrigation works, shifting patterns of land use, climate change and overfishing.[31] The fragmentation of river systems caused by dams is particularly damaging, as the massive structures impede fish migration, hinder larvae distribution and alter hydrology in ways that disturb breeding patterns.[32]

How far fish stocks will decline depends on various factors, including how many dams are ultimately built. One estimate indicates that by 2030, if all planned eleven dams are constructed on the lower Mekong mainstream, fisheries production would decline by 900,000 tonnes, or 41 per cent.[33] By some metrics, current and planned tributary dams, almost all of them in Laos, will have an even more severe impact on the supply and diversity of fish.[34] Fish catches on dammed Mekong tributaries – the Pak Mun, Sesan, Nam Theun and Hinboun Rivers – have fallen by 30 to 90 per cent, with rates concentrated toward the higher end of this range.[35] Some 74 fish species are endangered, with eighteen categorised as critically endangered, though gaps in data mean the actual numbers are likely higher.[36] A recent study of the largest fishery in

[30] "The Flow of the Mekong", Mekong River Commission, November 2009, pp. 8-9.
[31] R.M. Friend and R.I. Arthur, "Overplaying Overfishing: A Cautionary Tale from the Mekong", *Society & Natural Resources: An International Journal*, vol. 25, no. 3 (2012), p. 290.
[32] Baird and Hogan, "Hydropower Dam Development and Fish Biodiversity in the Mekong River Basin", op. cit., p. 1.
[33] Christopher D. Golden et al., "Impacts of Mainstream Hydropower Development on Fisheries and Human Nutrition in the Lower Mekong", *Frontiers in Sustainable Food Systems*, vol. 3, no. 93 (2019), p. 3.
[34] Guy Ziv et al., "Trading-off Fish Biodiversity, Food Security and Hydropower in the Mekong River Basin", *PNAS*, vol. 109, no. 15 (2012), p. 5610.
[35] Stuart Orr et al., "Dams on the Mekong River: Lost Fish Protein and the Implications for Land and Water Resources", *Global Environmental Change*, vol. 22 (2012), p. 926.
[36] According to the International Union for Conservation of Nature Red List of Threatened Species. Kathy Hughes, *The Mekong's Forgotten Fishes and the Emergency Recovery Plan to Save Them* (Gland, Switzerland, 2024), p. 8.

the lower Mekong basin, in Cambodia's Tonle Sap River, found an average decline of 90 per cent in the populations of 90 of 110 species between 2003 and 2019.[37]

### B. *Sediment, Erosion and Subsidence*

Dams and sand mining are reducing the amount of sediment in the Mekong and its movement downstream, which occurs primarily during the wet season. Reduced flows of sediment have extremely harmful effects on soil fertility, particularly in the floodplains and delta, imperilling the productivity of farms and threatening natural ecosystems, such as flooded forests that provide habitats for fish and other aquatic fauna.[38] Falls in sediment flows will also lead to greater riverbank erosion and accelerate land subsidence in the delta.

Mekong dams are trapping enormous quantities of sediment. The basin's sediment load comes mainly from the Mekong's upper reaches, the Lao highlands and the 3S basin; it travels downstream before being deposited in the Cambodian floodplains and the delta. If all the mainstream dams being planned are built, it is estimated that they could trap as much as 96 per cent of the total amount of sediment that would otherwise settle in the delta.[39]

The dams' impact is compounded by sand mining, which constitutes an existential threat to the Mekong Delta, causing riverbank erosion and collapse that put homes and infrastructure at risk.[40] The Vietnam Disaster and Dyke Management Authority estimates there are 585 erosion hotspots in the delta, spanning more than 740km, 87 of which are "extremely dangerous". Riverbank erosion is accelerating, with 145 cases registered in the first six months of 2023 – more than in all of 2022.[41]

Sand mining has also lowered the riverbed in both Cambodia and Vietnam, which could reduce water and sediment flows between the Mekong River and the Tonle Sap, further jeopardising the all-important flood pulse. The reduction in sediment flow attributed to sand mining imperils the delta, as deposits of river sand help shield the delta's coast from erosion.[42] Reduction in the flow of sand through the region also

[37] Mathieu Chevalier et al., "Long-term Data Show Alarming Decline of Majority of Fish Species in a Lower Mekong Basin Fishery", *Science of the Total Environment*, no. 891 (2023), p. 6.
[38] Alexander J. Horton et al., "The Cambodian Mekong Floodplain under Future Development Plans and Climate Change", *Natural Hazards and Earth System Sciences*, vol. 22 (2022), p. 979.
[39] See G.M. Kondolf et al., "Dams on the Mekong: Cumulative Sediment Starvation", *Water Resources Research*, vol. 50 (2014), p. 5158; and "The Council Study, Volume 1: Summary Report", Mekong River Commission, January 2018, p. 56. The Commission estimates that 35 per cent of sediment loss would be attributable to dams in China, 32 per cent to lower Mekong mainstream dams, 30 per cent to lower Mekong tributary dams and 3 per cent to "processes" in the Mekong Delta, presumably referring to sand mining. "The Council Study, Cumulative Impact Assessment, Key Findings Report", Mekong River Commission, December 2017, p. 50.
[40] Seminar comments by Marc Goichot of World Wildlife Federation at Foreign Correspondents' Club of Thailand, Bangkok, 30 November 2022. Christopher R. Hackney et al., "River Bank Instability from Unsustainable Sand Mining in the Lower Mekong River", *Nature Sustainability* (2020), p. 2.
[41] "Take from the river, the river takes back", *Vietnam Express*, 5 September 2023.
[42] Goichot et al., "Sand Mining and Climate Adaptation", op. cit., pp. 61-64.

results in increasing saltwater intrusion and the loss of coastal mangrove forests, which provide a crucial barrier against typhoons.[43]

### C. *Climate Change*

Climate change exacerbates the impact of environmental degradation caused by the exploitation of the Mekong and its tributaries. Scientists predict increased temperatures and annual precipitation across the region, prolonged droughts in the basin's south and east, increased flood depths and longer periods of inundation in the floodplain and delta, as well as greater saltwater intrusion in the delta.[44] Already, the lower Mekong has experienced severe droughts in 2009-2010, 2015-2016, 2019-2021 and 2023-2024. The combination of climate change, irrigation works and dams will homogenise the river's flows, which are likely to increase in the dry season and decrease in the wet season. One clear victim of these changes will be the annual flood pulse.[45] The flooded area in the Tonle Sap basin is expected to decline by 11 per cent by 2050, reducing sedimentation and aquatic primary production, namely the generation of the most basic parts of the food chain, by up to 59 per cent and 38 per cent, respectively. Soil fertility and fish stocks that sustain local livelihoods are expected to decline as a result.[46]

The Mekong region's governments have justified the wave of dam building as part of their campaign to mitigate climate change and create sustainable energy sources.[47] But proponents of hydropower, particularly in China and Laos, have gone further, arguing that climate change is the main driver of environmental problems in the basin and deflecting attention from the impact of dams.[48] Critics of dam-building are not impressed by these arguments. They respond by pointing to research showing that, aside from the very real impact of dams on the basin, free-flowing rivers serve as car-

[43] Christian Jordan et al., "Sand Mining in the Mekong Delta Revisited: Current Scales of Local Sediment Deficits", *Scientific Reports*, vol. 9 (2019), p. 9.
[44] Jaap Evers and Assela Pathirana, "Adaptation to Climate Change in the Mekong River Basin: Introduction to the Special Issue", *Climatic Change* (2018).
[45] Dry season flow is expected to increase up to 60 per cent, especially early in the dry season (March and April), and wet season flow to decrease by up to 25 per cent in July. Steven Ly, Takahiro Sayama and Sophal Try, "Integrated Impact Assessment of Climate Change and Hydropower Operation on Streamflow and Inundation in the Lower Mekong Basin", *Progress in Earth and Planetary Science*, vol. 10, no. 55 (2023), pp. 13-14.
[46] Horton et al., "The Cambodian Mekong Floodplain under Future Development Plans and Climate Change", op. cit., p. 968. Aquatic primary production is the process by which plants and algae convert sunlight, carbon dioxide and water into organic matter – the foundation of the food web – through photosynthesis.
[47] Mira Käkönen and Anja Nygren, "Resurgent Dams: Shifting Power Formations, Persistent Harms and Obscured Responsibilities", *Globalizations* (2022), pp. 8, 13.
[48] Crisis Group interview, hydropower expert, Chiang Mai, Thailand, January 2024. See also Hoang Thi Ha, "Can Mekong stingrays tell the Chinese dam story well?", *South China Morning Post*, 4 October 2022.

bon sinks that work to absorb greenhouse gases, while reservoirs of large dams emit methane, a damaging by-product of rotting vegetation, especially in tropical regions.[49]

Rather than providing an alternative energy source and part of the answer to climate change, dams may also be less sustainable than their proponents say. Climate change could reduce the region's hydropower production due to greater variability in streamflow, shifting seasonal flows and more evaporation from reservoirs. Scientists predict that, by 2100, climate change will reduce the region's hydropower capacity by between 5.9 per cent (in a "below 2°C" scenario) and 8.2 per cent (in an "above 4°C" scenario).[50]

Given the threats facing the delta from climate change and infrastructure development, Vietnam has been more active than its neighbours in advancing mitigation policies. Hanoi promulgated Mekong Delta Plans in 2013 and 2022, both prepared in partnership with the Netherlands. In 2017, following an extreme drought the year before, it issued Resolution 120 on Sustainable and Climate-Resilient Development of the Mekong Delta, which aims to promote adaptation to changing environmental conditions, including shifting from intensive rice cultivation toward higher-value produce and aquaculture, particularly shrimp farming, which is suited to brackish water.[51] Implementation of Resolution 120 has nevertheless been hampered by discord between upstream and downstream provincial governments.[52]

### D. *Human Security*

Current and future hydropower projects in the lower Mekong basin are expected to produce "serious and irreversible" environmental harm and loss of biodiversity, badly hurting the basin's ecosystem.[53] This damage, in turn, poses threats to human security, particularly to rural populations whose livelihoods are tied to fishing and farming.

Research on the costs and benefits of Mekong hydropower development has indicated that it is, in purely monetary terms, a losing proposition. In other words, the costs represented by the loss of fisheries, sediment and nutrients, as well as the investments needed to provide alternative livelihoods for those affected, are greater than the benefits derived from electricity generation, irrigation and flood control.[54]

[49] Marc Goichot, "Dammed if they don't: Mekong countries face crucial choice today", *The Nation*, 12 January 2016; Danny Marks and Jun Zhang, "Circuits of Power: Environmental Injustice from Bangkok's Shopping Malls to Laos' Hydropower Dams", *Asia Pacific Viewpoint* (2019), p. 303.
[50] In the "below 2°C" scenario, the rise in global annual mean temperature from pre-industrial times (1850-1900) to 2100 remains below 2°C. The "above 4°C" scenario projects a 4.3°C rise from the pre-industrial level by 2100. "Climate Impacts on South and Southeast Asian Hydropower", International Energy Agency, 2022, pp. 4, 27, 55.
[51] Crisis Group interview, Dang Kieu Nhan, director, Mekong Delta Development Research Institute, Can Tho University, Can Tho, Vietnam, 8 August 2022. According to a Western diplomat in Hanoi, however, "The [2022] integrated plan is high in aspiration but low in detail". Crisis Group interview, Hanoi, Vietnam, December 2022.
[52] Crisis Group interviews, academics, Ho Chi Minh City and Can Tho, Vietnam, August 2022.
[53] "MRC Strategic Environmental Assessment of Hydropower on the Mekong Mainstream: Summary of the Final Report", op. cit., p. 12.
[54] Apisom Intralawan et al., "Tradeoff Analysis between Electricity Generation and Ecosystem Services in the Lower Mekong Basin", *Ecosystem Services*, vol. 30 (2018), p. 27.

The costs and benefits are also unevenly distributed: the benefits of hydropower flow principally to electricity distributors, developers, financiers and governments, while the costs are borne mainly by poor, rural communities.[55] Environmental problems associated with large-scale hydropower also expose women who rely on rivers for subsistence and independent sources of income to particular harm.[56]

The fall-out of hydropower is felt most immediately by those compelled to resettle from near dams or reservoirs, which are often in remote areas populated by ethnic minorities. There are no reliable figures on how many people have been displaced by dams in the Mekong basin, but available information points already to tens of thousands, mostly in Laos.[57] A 2010 Mekong River Commission report estimated that almost 107,000 people from the lower basin countries would need resettlement if all eleven proposed mainstream dams in the area are built.[58] In general, relocation and compensation efforts for displaced people in the region have been inadequate.[59] While governments usually provide housing to the resettled, compensation for land and provision of livelihood opportunities is rarely commensurate with the homes and the jobs that have been lost. Money cannot make up for less tangible losses, like a sense of community.[60] Resettlement also tends to place additional burdens on women, including financial pressures and increasing household responsibilities.[61]

The starkest example of the threat posed by hydropower is the July 2018 collapse of a saddle dam of the Xe Pian-Xe Namnoy facility in southern Laos. According to official figures, the accident killed 71 people and displaced more than 6,000 others; another 5,000 people were uprooted in Cambodia.[62] There have been at least three earlier dam collapses in Laos.[63]

[55] "MRC Strategic Environmental Assessment of Hydropower on the Mekong Mainstream: Summary of the Final Report", op. cit., p. 11.
[56] "Gender and Water Governance in the Mekong Region", IUCN and Oxfam, 2018, p. 2.
[57] According to a 2013 study, the Mekong River Commission identified 69,413 people displaced by tributary dams in Laos, but its database is incomplete. Carl Middleton, "The Political Ecology of Large Hydropower Dams in the Mekong Basin: A Comprehensive Review", *Water Alternatives*, vol. 15, no. 2 (2022), p. 271.
[58] "MRC Strategic Environmental Assessment of Hydropower on the Mekong Mainstream: Summary of the Final Report", op. cit., p. 16.
[59] Middleton, "The Political Ecology of Large Hydropower Dams in the Mekong Basin: A Comprehensive Review", op. cit., pp. 271-272.
[60] Crisis Group interviews, villagers resettled from Lower Sesan 2 dam reservoir site, Kbal Romeas, Stung Treng, Cambodia, September 2022; villagers resettled from Xanamxai, Attapeu, Lao PDR, June 2023.
[61] Crisis Group interviews, villagers resettled from Lower Sesan 2 dam reservoir site, Kbal Romeas, Stung Treng, Cambodia, September 2022. See also Lisa Denney et al., "Inclusion in Decision-making around Transboundary Water Governance in the Mekong Sub-region", Institute for Human Security and Social Change, La Trobe University, July 2021, p. 17, note 10.
[62] Ian G. Baird, "Catastrophic and Slow Violence: Thinking about the Impacts of the Xe Namnoy Dam in Southern Laos", *Journal of Peasant Studies* (2020), pp. 1-2.
[63] These are the penstock failure of the 250 MW Xekaman 3 dam, in December 2016, also in the Sekong basin; the 12 MW Nam Ao Dam in northern Laos on September 2017; and the 1 MW Kaeng Khouan Dam in northern Laos in November 2019. Kanya Souksakoun, "Towards Sustainable Hydropower: Policy Implementation and Livelihood Transformation in the Sekong Basin, Laos", Ph.D. dissertation, Australian National University, 2022, p. 143.

The most damaging effects of large-scale infrastructure in the Mekong basin will be felt in Cambodia and Vietnam – the floodplain and the river's delta. Spread over 40,900 sq km, Vietnam's Mekong Delta constitutes 20 per cent of the nation's territory and is home to seventeen million people. It is Vietnam's most productive agricultural and fishing region, accounting for half of national rice production and 90 per cent of rice exports.[64]

Hydropower and sand mining are not the only threats to the delta. Its low mean elevation, just 0.8m above sea level, puts the region at risk of submergence as oceans rise due to climate change.[65] To make matters worse, the region is sinking as a result of land subsidence, caused by groundwater extraction and exacerbated by declining sediment deposit.[66] Land subsidence, which aggravates saltwater intrusion, is the most pressing threat, with the delta sinking at three to four times the rate of sea level rise.[67] At current rates, the sea will put most of the delta under water by 2100.[68] "It took 6,000 years to build the delta, but it can be destroyed by erosion in 200 years", lamented Nguyen Huu Thien, an environmental expert.[69]

By threatening rural livelihoods, large-scale infrastructure along the river also has consequences for food security. Tens of millions depend on the basin's fish in their diet.[70] But dams pose a threat to migratory fish, which account for 60 per cent of the fish catch in the lower Mekong basin and provide between 49 and 83 per cent of the animal protein people consume in Cambodia, Laos, Thailand and Vietnam.[71] One study estimated that the population at risk of inadequate protein intake could reach

[64] After unification in 1975, the government promoted intensification of rice cultivation to three crops per year, particularly in the upper delta. This practice made it necessary to build high dykes, which protect crops from flooding but also deprive the soil of sediment and natural pest control. The dykes, in turn, require greater use of groundwater, fertiliser and pesticides, reducing profits for farmers and further harming the environment. Crisis Group interviews, academics, Can Tho, Vietnam, August and December 2022. Today, Vietnam is the world's second largest rice exporter.

[65] P.S.J. Minderhoud et al., "Mekong Delta Much Lower than Previously Assumed in Sea-Level Rise Impact Assessments", *Nature Communications*, vol. 10, no. 1 (2019).

[66] The Intergovernmental Panel on Climate Change determined with high confidence that the effects of climate change, land subsidence and human activity will lead to "higher flood levels and prolonged inundation in the Mekong Delta". "Climate Change 2021: The Physical Science Basis. Contribution of Working Group I to the Sixth Assessment Report of the IPPC", Intergovernmental Panel on Climate Change, 2021, p. 139.

[67] Crisis Group interviews, Nguyen Huu Thien, ecologist, Can Tho, Vietnam, 7 August 2022; Dao Trong Tu, former member, National Mekong Committee, Hanoi, 9 December 2022.

[68] Crisis Group interview, Dao Trong Tu, former member, National Mekong Committee, Hanoi, 9 December 2022. See also "Vietnam's Mekong Delta, world's youngest river basin, might be first one gone", *VnExpress*, 4 September 2023; and R.J.P. Schmitt et al., "Losing Ground: Scenarios of Land Loss as Consequence of Shifting Sediment Budgets in the Mekong Delta", *Geomorphology*, no. 294 (2017).

[69] Crisis Group interview, Nguyen Huu Thien, ecologist, Can Tho, Vietnam, August 2022.

[70] R. Edward Grumbine, "Using Transboundary Environmental Security to Manage the Mekong River: China and South-East Asian Countries", *International Journal of Water Resources Development* (2017), p. 3.

[71] Ratha Sor et al., "Fish Biodiversity Declines with Dam Development in the Lower Mekong Basin", *Nature*, no. 13 (2023), p. 4.

2.23 million by 2030.[72] Cambodia's fisheries will be particularly hard-hit should all the proposed mainstream dams be constructed, accounting for 70 per cent of the total projected decline in fish catch.[73]

The impact of dams in the Mekong basin has already accelerated migration of rural people to urban areas, sometime across national borders.[74] Net outmigration can be seen in Vietnam's Mekong Delta, with more than one million, mostly young, people leaving between 2009 and 2019.[75] The drivers of this migration are complex, and the relative importance of pre-existing economic hardship and slow-burning environmental change is difficult to determine.[76] But given the growing precariousness of farming livelihoods in the delta, it is likely that environmental pressures are playing more of a role in decisions to move to the city.[77] In Cambodia, the declining fish catch in the Tonle Sap Lake has spurred many young people to migrate in search of wage labour.[78] This growing pool of young workers unable to subsist in traditional occupations may increase vulnerability to human trafficking and generate more recruits for the transnational criminal enterprises that are booming in mainland South East Asia.[79]

Large-scale displacement – or involuntary migration – due to environmental degradation is also looming in the lower Mekong, particularly in the delta. Together, rising sea levels, land subsidence and storm surges will displace between 3.3 million and 6.3 million people in the lower Mekong by 2050, according to the World Bank.[80] The submergence of swathes of the delta by 2100 would inevitably displace millions more.[81] Whether all these people could adapt to entirely new lives is, at best, uncertain.[82] For Vietnam, in particular, the prospect of lost territory, mass displacement and devastation of the country's most productive agricultural land would have life-

[72] Golden et al., "Impacts of Mainstream Hydropower Development on Fisheries and Human Nutrition in the Lower Mekong", op. cit., p. 1.
[73] Yuichiro Yoshida et al., "Impacts of Mainstream Hydropower Dams on Fisheries and Agriculture in Lower Mekong Basin", *Sustainability*, vol. 12, no. 2408 (2020), p. 16.
[74] Cambodia, Laos, Myanmar and Vietnam are mainly sending countries, and Thailand primarily a destination for economic migrants from the region. Viviane Clement et al., *Groundswell: Acting on Internal Climate Migration: Part 2* (Washington, 2021), p. 48.
[75] Crisis Group interview, migration expert, Ho Chi Minh City, August 2022. See also "Natural disasters cause people to flee Mekong Delta", *VnExpress*, 14 December 2020.
[76] Carl Middleton et al., "Climate Change, Mobility and Human Rights: 'Slow Onset' Environmental Change and Displacement in the Mekong Region", Center for Social Development Studies/Raoul Wallenberg Institute, March 2023.
[77] Crisis Group telephone interview, hydropower expert, June 2022. Crisis Group interviews, migration expert, Ho Chi Minh City, Vietnam, August 2022; environment expert, Can Tho, Vietnam, August 2022. See also Duong Phuong Le, Adriana Melgar and Adam Savelli, "The Implications of Climate-related Mobility for Human Security in Vietnam's Mekong Delta: A Scoping Review", CGIAR Initiative Report, 2022, p. 10.
[78] Crisis Group interviews, villagers, Kampong Chhnang, Cambodia, September 2022.
[79] Crisis Group Report, *Transnational Crime and Geopolitical Contestation along the Mekong*, op. cit.
[80] Clement et al., *Groundswell: Acting on Internal Climate Migration: Part 2*, op. cit., p. 42.
[81] P.S.J. Minderhoud et al., "Groundwater Extraction May Drown Mega-delta: Projections of Extraction-induced Subsidence and Elevation of the Mekong Delta for the 21st Century", *Environmental Research Communications*, vol. 2 (2020).
[82] Golden et al., "Impacts of Mainstream Hydropower Development on Fisheries and Human Nutrition in the Lower Mekong", op. cit., p. 2.

changing repercussions for millions of people.[83] It is hard to predict the full political and social implications of such sweeping, disruptive change with no apparent safety net, but at the very least it raises the spectre of rising social tensions in pockets of the Mekong basin and could well have wider destabilising implications.[84]

[83] Crisis Group interviews, hydropower expert, July 2022 academic, Can Tho, Vietnam, August 2022.
[84] Crisis Group interview, geographer, Chiang Rai, Thailand, January 2024. For a discussion of four hypothetical scenarios of climate-induced conflict and cooperation in the lower Mekong basin, see John Lichtefeld, "Climate Security in Mainland Southeast Asia: A Scenarios-Based Assessment", Council on Strategic Risks, November 2022.

## IV. Mekong Basin Governance

In many places, transboundary rivers spur cooperation rather than conflict between states.[85] The Mekong is a case in point. But even though the region's governments collaborate with one another over their shared river, they have failed to restrain infrastructure development or pay proper heed to the environmental harm that it has caused. Failure to protect the Mekong's ecosystem is in many regards a matter of choice by the region's governments. Thus far, local communities feel the strain of environmental damage more directly than states as a whole, fuelling tensions between localities and governments rather than between states.[86]

### A. *Institutions and Transboundary Governance*

There are three major institutional frameworks that promote infrastructure development in the Mekong basin: the Asian Development Bank's Greater Mekong Subregion Initiative (GMS), the Mekong River Commission (MRC) and China's Lancang-Mekong Cooperation (LMC) initiative.

Launched in 1992 with Japan's support, the GMS sought to realise a "peace dividend" by binding the region together through infrastructure development and investment after decades of war in Vietnam, Laos and Cambodia. With the Asian Development Bank serving as its secretariat, the GMS emphasised connectivity through transport infrastructure, but also encompassed agriculture, energy, telecommunications and trade liberalisation. The GMS concept was "informed by the liberal political theory that rather than being a cockpit of war, the economic transformation of the region will contribute to regional peace and order".[87]

In 1995, Cambodia, Laos, Thailand and Vietnam signed the Agreement on the Cooperation for Sustainable Development of the Mekong River Basin, establishing the MRC as the region's only treaty-based river basin organisation.[88] The agreement com-

[85] Aaron T. Wolf, "Shared Waters: Conflict and Cooperation", *Annual Review of Environment and Resources*, vol. 32 (2007), pp. 241-269.

[86] Laohasiriwong and Oishi, "Managing the Mekong River Conflicts: Political Stability at the Cost of Local Communities", op. cit., p. 157.

[87] Donald E. Weatherbee, "Cooperation and Conflict in the Mekong River Basin", *Studies in Conflict and Terrorism*, vol. 20, no. 2 (1997), p. 168. The GMS covers a geographic area much broader than the Mekong basin, including all five mainland South East Asian countries as well as China's Yunnan province and Guangxi Zhuang autonomous province, the latter of which lies entirely outside the basin. Yunnan joined the GMS in 1995, marking China's first foray into a regional multilateral arrangement; Guangxi joined in 2005. Suzanne Ogden, "The Impact of China's Dams on the Mekong River Basin: Governance, Sustainable Development and the Energy-Water Nexus", *Journal of Contemporary China*, vol. 32, no. 139 (2023), p. 155.

[88] The MRC's lineage stretches back to the formation of the Mekong Committee for Coordination of Investigations of the Lower Mekong in 1957, established by Cambodia, Laos, Thailand and the Republic of Vietnam, with help from the UN and the U.S., to coordinate planning and investigation of water resources development. It carried out surveys of the Mekong basin, but wars in Indochina prevented it from carrying out large-scale hydropower plans. Cambodia withdrew from the Committee under the Khmer Rouge in 1975, at which point Thailand, Laos and Vietnam established the Interim Mekong Committee. In 1991, Cambodia applied for readmission to the committee, prompting negotiations that led to the 1995 Mekong Agreement.

mits the member countries to cooperate in sustainable development of water and related resources in the interests of "an economically prosperous, socially just and environmentally sound river basin".[89] The Commission consists of a ministerial-level Council, a Joint Committee staffed by senior member state officials responsible for executing Council decisions, and the Secretariat, headquartered in Vientiane, providing administrative and technical services. Phnom Penh hosts the MRC's Regional Flood and Drought Management Center. National Mekong Committees, usually under the auspices of water and resources ministries, coordinate between their respective line ministries and the MRC.

The MRC Joint Committee is mandated to prepare a basin development strategy based on the principles of Integrated Water Resources Management, which aim "to guide all actors involved in Mekong water-related issues toward achieving improvements in the environmental, social and economic state of the Mekong River Basin".[90] Critics describe Integrated Water Resources Management as so "vague and generic" that it constitutes an "empty motto".[91]

Tensions between promoting economic development and managing the basin's resources have regularly surfaced in the MRC, often pitting Western donors interested in environmental preservation against member states preoccupied with the economic benefits of water use.[92] The 1995 Mekong Agreement stressed the need for equitable sharing of water and maintenance of minimum flows, since these issues are potential causes of interstate conflict that could threaten regional stability. But it provided no effective means of preserving the river system's ecosystem, while the interests of the basin's most vulnerable people are rarely considered in policymaking.[93] Since 2016, the MRC has also been undergoing a process of "riparianisation", meaning that mem-

[89] "Agreement on the Cooperation for Sustainable Development of the Mekong River Basin", 5 April 1995. Of note are the injunctions "to utilize the waters of the Mekong River system in a reasonable and equitable manner in their respective territories" (Article 5) and "To make every effort to avoid, minimize and mitigate harmful effects that might occur to the environment … from the development and use of the Mekong River Basin water resource" (Article 7).

[90] "The Integrated Water Resources Management–based Basin Development Strategy for the Lower Mekong Basin 2021-2030 and the MRC Strategic Plan 2021-2025", Mekong River Commission, 2021, p. 3. Integrated Water Resources Management is "a process that promotes the coordinated development and management of water, land and related resources, in order to maximise economic and social welfare in a balanced way without compromising the sustainability of the ecosystems". "Mekong Basin Planning: The Basin Development Plan Story", Mekong River Commission, 2013, p. 31.

[91] Masao Imamura, "Introduction: Water Governance in the Mekong Region" in Louis Lebel et al. (eds.), *Democratizing Water Governance in the Mekong Region* (Bangkok, 2007), pp. 1-2.

[92] Claudia Kuenzer et al., "Understanding the Impact of Hydropower Developments in the Context of Upstream-Downstream Relations in the Mekong River Basin", *Sustainable Science*, vol. 8 (2013), p. 580.

[93] Crisis Group telephone interview, hydropower expert, July 2022. See also Chris Sneddon and Coleen Fox, "Rethinking Transboundary Waters: A Critical Hydropolitics of the Mekong Basin", *Political Geography*, vol. 25 (2006), p. 192; and Denney et al., "Inclusion in Decision-making around Transboundary Water Governance in the Mekong Sub-region", op. cit., p. 1. Gender inequalities in the region mean that women have fewer opportunities to participate in formal decision-making processes on water resources. Karen Delfau and Pichamon Yeophantong, *State of Knowledge: Women and Rivers in the Mekong Region* (Oakland, 2020), pp. 18-19.

ber states rather than benefactors from outside the region are taking on greater responsibility for funding and leadership.[94]

The MRC is often reproached for failing to curb rampant hydropower development or even to ensure that members states adhere to their obligations.[95] Such criticism minimises the extent to which the MRC is "a governed rather than governing organization".[96] As the MRC's chief executive Anoulak Kittikhoun said, "Some people have a mission to point out the culprits. We do not have that mission".[97] Member states' National Mekong Committees have little influence on energy policy. Moreover, water and resources ministries are invariably weaker than energy ministries that oversee hydropower.[98] The MRC's effectiveness is also constrained by the fact China and Myanmar are not members, though both are associated as dialogue partners.[99] China declined to join, concerned about its sovereignty and suspicious of the MRC's dependence on Western funding.[100]

Beijing, for its part, established the Lancang-Mekong Cooperation initiative in 2015 as an extension of its Belt and Road Initiative, at least partly in response to criticism of its stewardship of the Lancang/Mekong.[101] All six basin countries are members. The Lancang-Mekong Cooperation framework has three pillars: political and security issues; economics and sustainable development; and social, cultural and people-to-people exchanges.[102] It is the only Mekong region cooperative framework to encom-

[94] Member states aim to fund the MRC's core functions fully by 2030. For 2022, members contributed $4.5 million and foreign donors $4.2 million. "FS MRC Consolidated Report, Audit", Mekong River Commission Secretariat, 31 December 2022, p. 13; Luke Hunt, "Mekong River Commission faces radical change", *The Diplomat*, 22 January 2016.

[95] Milton Osborne, "How the Mekong River Commission ignores reality", The Interpreter (blog), 4 December 2022; Brian Eyler, "Mekong River agency must step up", *Bangkok Post*, 1 April 2023. The 1995 Agreement is legally binding on members but lacks enforcement mechanisms. Oliver Hensengerth, "Where is the Power? Transnational Networks, Authority and the Dispute over the Xayaburi Dam on the Lower Mekong Mainstream", *Water International*, vol. 40, no. 5-6, (2015), p. 917; Jo Toriyabe, "The Prevention and Resolution of International Conflicts over Transboundary Water Resources in the Mekong River Basin", 摂南法学第58 号 (2021), p. 21.

[96] Philip Hirsch, "IWRM as a Participatory Governance Framework for the Mekong River Basin?" in Stina Hansson et al. (eds.), *Politics and Development in a Transboundary Watershed: The Case of the Lower Mekong Basin* (Dordrecht, 2012), p. 158.

[97] Aron White, "Pointing out culprits of Mekong's problems is not our mission: MRC Secretariat CEO", The Third Pole, 10 April 2023.

[98] Carl Middleton and John Dore, "Transboundary Water and Electricity Governance in Mainland Southeast Asia: Linkages, Disjunctures and Implications", *International Journal of Water Governance*, no. 1 (2015), p. 111; Hensengerth, "Where is the Power?", op. cit., p. 917.

[99] China participates in the MRC's international conferences, annual dialogue meetings and summits every four years.

[100] Crisis Group interview, diplomat, Bangkok, December 2022.

[101] Chinese Premier Li Keqiang proposed the LMC at the 17th ASEAN-China Summit in November 2014. It was based on a 2012 Thai proposal for regional cooperation in sustainable development in the Mekong basin. The first LMC foreign ministers' meeting took place on 12 November 2015 in Jinghong, China. The first leaders' meeting was held in Sanya, China, on 23 March 2016.

[102] Supporting these pillars are five sub-themes: connectivity, production capacity, cross-border economic cooperation, water resources, and agriculture and poverty reduction.

pass political and security issues.[103] The Lancang-Mekong Water Resources Cooperation Centre was established in Beijing in 2017 to sponsor research and facilitate capacity building and technical exchange.

Well-funded, the LMC emphasises concrete outcomes, following "a government-guided, multiple-participation and project-oriented model".[104] By facilitating China-funded projects in member countries, it has helped Beijing align itself with downstream governing elites.[105] But rather than a multilateral forum to address basin-wide planning, the LMC is an umbrella for bilateral agreements between China and other members, excluding outside powers.[106]

### B. *Prior Consultation and Impact Assessments*

In 2003, the Mekong River Commission Council adopted the Procedures for Notification, Prior Consultation and Agreement, negotiated by the member states in accord with the 1995 Mekong Agreement. The procedures are designed for concerned member states to assess the potential transboundary impact on ecosystems and livelihoods of "any use of water which may have a significant impact to the water quality or flows regime of the mainstream of the Mekong River System by any member State", and to devise measures to mitigate any harm.[107] For proposed projects on the river's mainstream, the procedures involve a six-month process of evaluation and formal consultations among member states meant to conclude with an agreement by the MRC Joint Committee on the conditions under which the project will proceed, although consensus is not mandatory.[108] As for proposed developments on tributaries within each country's own territory, member states must only notify the others of their plans.

The first test of the procedures – and the first illustration of their weakness – came in the shape of the Xayaburi Dam, a Thai-financed project to generate 1,285MW, 95 per cent of which was to be exported to Thailand.[109] In September 2010, the Lao National Mekong Committee submitted documents to the MRC Secretariat, initiating the six-month process. But the Joint Committee meeting that was to conclude it

[103] Pongphisoot Busbarat et al., "China and Mekong Regionalism: A Reappraisal of the Formation of Lancang-Mekong Cooperation", *Asian Politics & Policy*, no. 13 (2021), p. 202.
[104] Sanya Declaration of the First Lancang-Mekong Cooperation Leaders' Meeting, 23 March 2016.
[105] Ogden, "The Impact of China's Dams on the Mekong River Basin", op. cit., pp. 161-162.
[106] Toriyabe, "The Prevention and Resolution of International Conflicts over Transboundary Water Resources in the Mekong River Basin", op. cit., p. 18; Malcolm F. McPherson, "China's Role in Promoting Transboundary Resource Management in the Greater Mekong Basin", Ash Center for Democratic Governance and Innovation, March 2020, p. 14; Shang-su Wu, "Lancang-Mekong Cooperation: The Current State of China's Hydro-politics", in Bhubhindar Singh and Sarah Teo (eds.), *Minilateralism in the Indo-Pacific: The Quadrilateral Security Dialogue, Lancang-Mekong Cooperation Mechanism and ASEAN* (London, 2020), p. 80.
[107] Mekong River Commission, "Procedures for Notification, Prior Consultation and Agreement", 2003.
[108] If the Joint Committee does not reach agreement, the proposal is put before the Mekong Council. If the Council fails to reach agreement, the issue is referred to "normal diplomatic channels", ie, consigned to the member states. "Procedures for Notification, Prior Consultation and Agreement", Mekong River Commission, December 2016, pp. 3-5.
[109] "MRC Strategic Environmental Assessment of Hydropower on the Mekong Mainstream: Summary of the Final Report", op. cit., p. 7.

in April 2011 failed to reach agreement and referred the matter to the MRC Council meeting in December. Vietnam then called the environmental impact assessment inadequate and urged that mainstream dams be deferred for a decade.[110] Cambodia also requested an extension of the consultation period.[111] Meanwhile, the Lao deputy minister for energy and mines argued that the process had concluded in April, underlining that it required only consultation, not agreement. In fact, dam construction had begun as early as late 2010.[112] In the end, the process did result in minor changes to the dam's design, but it concluded without member-state agreement, and Laos acted unilaterally to build the dam.[113]

The failures of the Xayaburi consultations led to reform of the procedures, including more timely release of information, introduction of regional stakeholder forums, beginning in 2017 for the Pak Beng dam, and release of Joint Action Plans to monitor design, construction and operation of dams after the consultations.[114] Despite these changes, many activists still regard the process as inadequate and lacking in transparency; some have boycotted consultations they perceive as window dressing.[115]

Hydropower planning is also undermined by poor-quality environmental impact assessments and the absence of mandatory transboundary environmental assessments. These studies inform government decisions about how to mitigate adverse environmental and social impact as well as how to provide compensation to affected communities. Commissioned by investors, these assessments too often lack rigour, are limited in geographic scale (sometimes taking into account effects no farther than a few kilometres from the dam) and time, and they proceed with little or no baseline information. The assessments also tend to be undertaken late in the planning cycle, after high-level decisions have been made, and fail to account for the cumulative transboundary impact of multiple dams.[116] Worse, sections of the assessment are

[110] "Mekong River Commission, Procedures for Notification, Prior Consultation and Agreement, Form for Reply to Prior Consultation", Viet Nam National Mekong Committee, 15 April 2011. An environmental assessment prepared for the Mekong River Commission recommended a ten-year moratorium on mainstream dams and commissioning of 50 studies on potential environmental impacts. International Centre for Environmental Management, op. cit.

[111] "Mekong River Commission, Procedures for Notification, Prior Consultation and Agreement Form/Format for Reply to Prior Consultation", Cambodia National Mekong Committee, 13 April 2011.

[112] Naho Mirumachi, *Transboundary Water Politics in the Developing World* (Abingdon, 2015), p. 126; Toriyabe, "The Prevention and Resolution of International Conflicts over Transboundary Water Resources in the Mekong River Basin", op. cit., pp. 17, 24.

[113] The modifications included installation of fish ladders, the effectiveness of which is unproven. Supalak Ganjanakhundee, "The tussle with China in the Mekong Basin", *Straits Times*, 17 November 2018.

[114] Oliver Hensengerth, "Inclusive Governance of Hydropower on Shared Rivers? Toward an International Legal Geography of the Lower Mekong Basin", *Frontiers in Climate* (2024), p. 7.

[115] Crisis Group interview, environmental expert, Chiang Mai, Thailand, January 2024. See also Save the Mekong Coalition, "Boycott of the Pak Lay Dam Prior Consultation Process", 24 August 2018; and Paritta Wangkiat, "Bali water forum discusses managing the Mekong", Mekong Eye, 27 May 2024. The Joint Action Plan for Pak Beng, for example, was not approved until two years after the agreement was concluded. Hensengerth, "Inclusive Governance", op. cit., p. 8.

[116] Elisabeth Jackson et al., "Inclusion in Decision-making around Transboundary Water Governance in the Mekong Sub-region: Cambodia Country Analysis", Institute for Human Security and Social Change, La Trobe University, July 2021, p. 15; Philip Hirsch, "Scaling the Environmental

frequently copied from earlier studies of other dams.[117] Referring to the Xayaburi and Pak Beng assessments, an academic from the Mekong Delta said: "If my undergraduate students turned in reports like that, they'd fail".[118]

### C. *Activism and Policymaking*

Governments in the region provide few channels through which citizens can register their needs and preferences with respect to large-scale infrastructure development. Civil society and environmental activists in the lower Mekong basin have challenged hydropower and other infrastructure development, but state clampdowns have further restricted the scope for activism over the past decade, especially in Laos, Cambodia and Vietnam.

In Laos, a one-party communist state, what little space civil society had to challenge the status quo dissipated after the enforced disappearance of Sombath Somphone, an educator and community development worker, abducted at a police checkpoint in Vientiane in December 2012.[119] Similarly, in Cambodia, space for activism has contracted discernibly after introduction of a new law on non-governmental organisations in 2010. Western donors faced greater obstacles in supporting local NGOs, which the government often portray as a threat to national sovereignty.[120] Court cases have hamstrung many environmental NGOs that challenge vested interests.[121]

In Vietnam, environmentalists who helped propel the beginning of a transition to clean energy have also come under government pressure. Since 2021, authorities

Commons: Broadening Our Frame of Reference for Transboundary Governance in Southeast Asia", *Asia Pacific Viewpoint* (2020), p. 9.

[117] David Blake and Pichamon Yeophantong, "Inclusion in Decision-making around Transboundary Water Resources in the Mekong Sub-region: Laos PDR Country Analysis", Institute for Human Security and Social Change, La Trobe University, July 2021, p. 12. The assessment for the Pak Lay dam in Laos copied extensively from a previous report on Pak Beng dam, which is located some 200km downstream. Similarly, the Sanakham dam assessment plagiarised that of Pak Beng. Pratch Rujivanarom, "Assessment reports of Pak Lay and Pak Beng dam found to be identical", *The Nation* (Thailand), 21 September 2018; "Geopolitics plays out on the Mekong with doubts on dams and promises of cooperation", ASEAN Today, 5 December 2020.

[118] Crisis Group interview, academic, Can Tho, Vietnam, August 2022. The MRC published "Guidelines for Transboundary Environmental Impact Assessment in the Lower Mekong Basin" in 2023, but responsibility for setting and enforcing standards remains with member states.

[119] In August 2019, Lao activist Od Sayavong disappeared from Bangkok. Houayheuang Xayabouly, who criticised the government's response to a dam collapse in 2018, was sentenced to five years in prison in November 2019. Another Lao activist, Anousa "Jack" Luangsuphom, was shot and wounded in Vientiane on 29 April 2023; the government attributed the attack to a business dispute, but it is widely believed he fell afoul of the authorities by criticising China's growing influence in Laos. On 16 May 2023, a Lao pro-democracy activist was found murdered in Thailand's Ubon Ratchathani province. "Laos activist Od Sayavong still missing six months on, while critics continue to be silenced", Civicus Monitor, 21 February 2020; "Vanished, shot, murdered: Laos activists spooked by spate of incidents", Agence France-Presse, 8 June 2023.

[120] W. Nathan Green and Ian G. Baird, "The Contentious Politics of Hydropower Dam Impact Assessments in the Mekong River Basin", *Political Geography*, vol. 83 (2020), p. 10.

[121] Crisis Group interview, environmental expert, Phnom Penh, September 2022. Mech Dara and George Wright, "The Cambodian activists risking their freedom for the environment", BBC, 9 April 2022; "Court bars Mother Nature activists from leaving the country to accept international award", CamboJA, 3 October 2023.

have detained and charged at least six such activists, most for tax evasion; two have since been released. The crackdown accelerated after Vietnam signed a Just Energy Transition Partnership with the G7 countries, Denmark and Norway, in December 2022, and upgraded ties with the U.S. in September 2023.[122] While Vietnamese authorities maintain the charges are legitimate law enforcement measures, human rights groups say the government regards civil society monitoring of its actions as an intolerable challenge.[123]

Environmental activism has had more clout in Thailand, but its effect has been to push Thai private investments into neighbouring countries where civil society is weaker, particularly Laos; in 2022, Thailand accounted for 85 per cent of Lao electricity exports.[124] Critics refer to Thailand's resource extraction as "the export of environmental injustices across borders".[125] Supported by international partners, Thai activists have attempted to thwart transnational projects through public campaigns, but their efforts have not yet succeeded in reversing decisions to build dams outside the country. In 2012, the Rak Chiang Khong Group, which represents a riparian community in Thailand, also filed suit against the state-owned Electricity Generating Authority, arguing that the Xayaburi Power Purchase Agreement – which makes the dam financially viable – would wreck local livelihoods that rely on fishing, riverbank gardens and harvest of an edible weed known as *kai*. Even so, after ten years, a Thai court dismissed the case, reasoning that any adverse impact would not be caused by the purchase agreement itself.[126]

Public pressure rooted in environmental concerns nevertheless appeared strong enough to push Thai authorities to reconsider agreements to buy electricity from proposed Lao dams on the Mekong mainstream.[127] In 2018, the Electricity Generating Authority of Thailand announced that it would defer a decision on a power purchase agreement for the Pak Beng dam until the review of Thailand's Power Development Plan was completed.[128] The secretary-general of Thailand's Office of National Water Resources said data supplied by developers of the proposed Sanakham dam was incomplete, and threatened to block a purchase agreement, citing national security concerns stemming from the possible impact on the position of the Thai-Lao

[122] Jonathan Head, "Vietnam jails fifth climate activist on tax charges", BBC, 28 September 2023; "Vietnam detains energy think-tank chief in latest arrest of environmental expert", Agence France-Presse, 1 October 2023; Rebecca Ratcliffe, "'They challenged the Communist monopoly': Vietnam regime turns on its climate champions", *The Guardian*, 17 October 2023.

[123] "Vietnam: 'The government hates losing face and has zero tolerance for criticism'", Civicus, 29 September 2023; "Vietnam's arrest of environmentalists draws fire amid surge of funding for green transition", Voice of America, 8 October 2023.

[124] "Electricity in Laos", Observatory of Economic Complexity.

[125] David J.H. Blake and Keith Barney, "Impounded Rivers, Compounded Injustice: Contesting the Social Impacts of Hydraulic Development in Laos", *International Journal of Water Resources Development* (2021), p. 5.

[126] "Thai court dismisses Mekong River activists' case against Xayaburi dam in Laos", Benar News, 17 August 2023.

[127] "Tide turning for Mekong protesters", *Bangkok Post*, 7 December 2020.

[128] "Thailand delays decision on power purchase from Pak Beng Dam", press release, International Rivers, 13 March 2018. Release of the power development plan has been repeatedly delayed amid new emissions targets, the COVID-19 pandemic and a surge in energy prices. "New power plan nears completion", *Bangkok Post*, 17 February 2024.

border.[129] At the time of publication, a power purchase agreement for Sanakham had yet to be concluded.

Electricity demand, meanwhile, is not the sole reason behind dam construction in the Mekong basin, and other interests at stake may be indifferent to public opposition.[130] State officials may be motivated by the desire to meet investment or growth targets, or to aggrandise particular government agencies.[131] Dam construction also affords opportunities for graft.[132] A hydropower expert observed that Laos' dam-building boom is driven more by the construction industry than the energy sector, while a geographer who studied Lao hydropower estimates that corruption accounts for 5 to 20 per cent of project costs for the country's dams.[133]

### D. *Project Finance and Power Trade*

Readiness to consider the environmental effects of dams and stand accountable for them is also undermined by the financial model used to create and run them. Beginning in the 1990s, Lao hydropower dams built primarily to export electricity relied on the "build-operate-transfer" model of financing. This model was first introduced by the multilateral development banks (World Bank and Asian Development Bank) and Western donors, who pitched dams as a fast route to economic development and poverty alleviation. These public-private partnerships typically involve a consortium of private investors and formation of an independent power producer company that builds and operates a dam for a concessionary period, usually 25 to 40 years, before the dam is transferred to the government.

As knowledge of the environmental harm caused by dams spread in the 2000s, multilateral development banks withdrew from sponsoring large hydropower projects and "non-traditional" players, particularly banks from the region (Thailand, Vietnam, Malaysia, Korea, China), stepped in to provide funding, looking for profits.[134]

[129] "Thailand threatens to reject electricity from Laos dam over environmental, security concerns", Benar News, 3 December 2020; "Thailand rejects new technical report on planned large Lao Mekong mainstream dam", Radio Free Asia, 24 January 2021.

[130] Crisis Group interviews, hydropower expert, July 2022; Cambodian political scientist, Phnom Penh, September 2022.

[131] Denney et al., "Inclusion in Decision-making around Transboundary Water Governance in the Mekong Sub-region", op. cit., p. 9.

[132] Crisis Group interview, hydropower expert, Chiang Mai, Thailand, January 2024. See also Danzig Sopera, "Corruption and Water Governance in the Mekong River Basin", Chr. Michelsen Institute, December 2022.

[133] Crisis Group telephone interview, hydropower expert, July 2022. See also Nathaniel Matthews, "Water Grabbing in the Mekong Basin: An Analysis of the Winners and Losers of Thailand's Hydropower Development in Lao PDR", *Water Alternatives*, vol. 5, no. 2 (2012), p. 402; and Nathaniel Matthews, "Drivers and Enablers of Hydropower Development in the Lao PDR: The Political Ecology of Mekong Riparians, Investors and the Environment", Ph.D. dissertation, King's College London, 2013.

[134] Merme et al., "Private Equity, Public Affair: Hydropower Financing in the Mekong Basin", op. cit., p. 21; Pon Souvannaseng, "Liquidated: US/Japan-Chinese Rivalry, Financial Crises and Explaining Shifts in Hydropower Finance Regimes in the Mekong", *Asian Perspective*, vol. 46, no. 1 (2022). There has since been another shift to a simpler model of finance through export credit agencies, particularly from China Exim Bank. Judith Plummer Braeckman et al., "Mapping the Evolving Com-

For Thai banks, for example, the energy sector offers potentially higher and faster returns than retail banking.[135] Lao authorities have every incentive to speed up approvals for dams to generate desperately needed revenue.[136] Large-scale hydropower is also driven by the need for importers to meet the European Union's Carbon Border Adjustment Mechanism standards on carbon emissions, and avoid penalties of €10-50 per tonne of carbon emissions resulting from production of imported goods.[137] The mechanism aims to eliminate the cost advantage of producing goods in countries with lax emissions standards.

During the concessionary period (typically 25-30 years), dams are primarily private rather than national assets, and investors' search for profit drives decision-making. Investors are shielded from accountability with respect to environmental, social and governance safeguards, responsibility for which falls upon governments that have limited capacity to enforce rules and large incentives to keep open the flow of profit from dams. The contracts governing the projects are not public, undermining transparency, and potential conflicts of interest are rife, as the host government is often an equity shareholder as well as a regulator, undermining its impartiality and ability to act in the public interest.[138] By the time they are transferred to the state, these dams have trapped significant amounts of sediment, meaning that they are less efficient, less capable of storing water and potentially less safe. Thus, they often require repairs. A former Cambodian minister said, "Dams need standards. We can't leave our destiny to investors".[139]

Much of the impetus behind building dams comes from companies that invest in them to supply consumers with electricity. Most Lao hydropower dams would not be built if Thailand and Vietnam did not agree to purchase the electricity.[140] Despite public campaigns in Thailand to curb electricity purchases from Lao dams, as well as the fact that Thailand has a large power surplus, the Electricity Generating Authority of Thailand continues to make contracts to buy hydropower from Lao mainstream

plexity of Large Hydropower Project Finance in Low and Lower-Middle Income Countries", University of Manchester and University of Cambridge Institute for Sustainability Leadership, 2020, p. 4.

[135] Pon Souvannaseng, "Fast Finance and the Political Economy of Catastrophic Dam Collapse in Lao PDR: The Case of Xe Pian-Xe Namnoy", *Pacific Affairs*, vol. 97, no. 2 (June 2024), p. 274.

[136] Ibid., pp. 275-276.

[137] "EU importers face low fine for failing to report emissions in trial phase", *Financial Times*, 17 August 2023.

[138] Merme et al., "Private Equity, Public Affair: Hydropower Financing in the Mekong Basin", op. cit., p. 27. Ame Trandem, "Offloading Risks & Avoiding Liabilities: How Financial Institutions Consider Hydropower Risks in Laos", *Focus on the Global South*, February 2019, pp. 11-12, 17.

[139] Crisis Group interview, Ambassador Pou Sothirak, executive director, Cambodian Institute for Cooperation and Peace, Phnom Penh, 14 September 2022.

[140] "MRC Strategic Environmental Assessment of Hydropower on the Mekong Mainstream: Summary of the Final Report", op. cit., p. 8. Thailand's National Energy Policy Committee agreed in 2021 to purchase 10,500MW of electricity (mostly hydroelectricity) annually from Laos. Vietnam is the third largest export market for Lao electricity, behind Thailand and Cambodia. In January, state-owned Vietnam Electricity entered into nineteen power purchase agreements for some 2,689MW of electricity from Lao hydropower plants. Piyaporn W., "Mekong resident and local agencies left in the dark about Pak Beng on the lower Mekong", *Bangkok Tribune*, 10 October 2022; "Vietnam inks deals to purchase electricity from 26 Lao hydropower plants", *Tuoi Tre News*, 8 January 2024.

Mekong dams.[141] The authority has regularly overestimated peak demand, resulting in excessive investment in new power plants to produce electricity that Thailand does not need.[142] The explanation is that under Thailand's cost-plus tariff system, the generating authority passes on the costs of overinvestment to consumers.[143] "They only care about money. I feel very disappointed that the authorities have failed to … protect the people", said Niwat Roikaew, an environmental activist.[144]

[141] These include power purchase agreements for the Luang Prabang dam with CK Power (17 November 2022), the Pak Lay dam with Gulf Energy Development (20 March 2023) and the Pak Beng dam (13 September 2023), also with Gulf. The Thai Energy Policy Council reportedly approved the Pak Beng agreement after Gulf Energy increased its stake in the dam project. Crisis Group interview, hydropower expert, Chiang Rai, January 2024. Pianporn Deetes and Gary Lee, "Pak Lay and Luang Prabang dam Power Purchase Agreements: Who benefits?", International Rivers, 30 March 2023; "EGAT secretly signed Pak Beng dam's Power Purchase Agreement", Transborder News, 17 September 2023.

[142] Supakhit Nuntavorakarn, "Thailand's Power Development Planning and a Just Energy Transition", in Carl Middleton and Tarek Ketelson (eds.), *Pathways to a Sustainable and Just Transformation of the Mekong Region's Electricity Sector* (Bangkok, 2022), pp. 127-130.

[143] Marks and Zhang, "Circuits of Power: Environmental Injustice from Bangkok's Shopping Malls to Laos' Hydropower Dams", op. cit., p. 301. Thai electricity consumers are subject to a fuel adjustment charge, originally called a "float time" or Ft charge, adjusted every four months by the Energy Regulatory Commission. "The Ft initially accounted for fuel price risk but grew in scope to include costs of new capacity, take-or-pay gas contracts, revenue shortfalls due to inaccurate demand forecasts, and foreign exchange risks". Chuenchom Sangarasri Greacen and Chris Greacen, "Thailand's Electricity Reforms: Privatization of Benefits and Socialization of Costs and Risks", *Pacific Affairs*, vol. 77, no. 3 (2004), p. 530.

[144] "EGAT secretly signed Pak Beng dam's Power Purchase Agreement", Transborder News, 17 September 2023.

## V. Geopolitics in the Mekong Basin

The Mekong region is an arena for global competition between China and the United States, in which both countries are seeking to expand their influence.[145] From Beijing's perspective, the lower Mekong region is akin to its backyard, a place where its clout should by rights be commensurate with its status as a major power.[146] The U.S. footprint in the region dates to anti-communist struggles beginning in the 1950s; for Washington and many of its allies and partners, contemporary engagement with the lower Mekong basin aims in large part to balance Beijing's sway.[147]

### A. *China and the U.S.*

Water and infrastructure are important facets of Sino-U.S. competition. The profusion of international Mekong-branded frameworks and initiatives derives from the region's perceived geopolitical importance and external powers' pursuit of influence there.[148] That said, these agreements are all geared toward promoting economic development and trade, notwithstanding programs aimed at energy transition and environmental sustainability.[149]

In 2009, the U.S. established the Lower Mekong Initiative, upgraded to the Mekong-U.S. Partnership in 2020, which promotes a "cautionary approach" to hydropower.[150] The Partnership emphasises "soft infrastructure" to support lower Mekong countries' "autonomy, economic independence, good governance and sustainable growth" through programs on development cooperation, transboundary water and natural resources management, non-traditional security, energy and infrastructure.[151] The Partnership also aims to keep other extra-regional powers engaged in the region.[152]

[145] Crisis Group Report, *Transnational Crime and Geopolitical Contestation along the Mekong*, op. cit., pp. 17-24.
[146] Busbarat et al., "China and Mekong Regionalism: A Reappraisal of the Formation of Lancang-Mekong Cooperation", op. cit., p. 204.
[147] The Trump administration labelled China a "strategic competitor", while the Biden administration has called China "America's most consequential geopolitical challenge". The White House, "National Security Strategy of the United States of America", December 2017; "National Security Strategy", October 2022.
[148] There are some thirteen Mekong-branded intergovernmental cooperation frameworks, including those sponsored by China, India, Japan, South Korea and the U.S. Benjamin Zawacki, "Implications of a Crowded Field: Sub-regional Architecture in ACMECS Member States", The Asia Foundation, 2019.
[149] Crisis Group interview, hydropower expert, Phnom Penh, September 2022. See also Sim Vreak, "Mapping Mekong Cooperation Complementarities and Policy Implications", AVI Perspective, 28 June 2019.
[150] The partnership involves the U.S., Thailand, Laos, Cambodia and Vietnam. Philip Hirsch, "The Shifting Regional Geopolitics of Mekong Dams", *Political Geography*, vol. 51 (2016), p. 72.
[151] "Launch of the Mekong-U.S. Partnership: Expanding U.S. Engagement with the Mekong Region", Office of the Spokesperson, U.S. Department of State, 14 September 2020.
[152] Under the Mekong-U.S. Partnership, the U.S. State Department supports the Mekong Water Data Initiative, Mekong Dam Monitor and Mekong Infrastructure Tracker, managed by the Stimson Center, which provide open-source data on developments in the region. It also supports USAID Mekong Safeguards, which provides "information and tools to apply environmental, social and governance

Since 2011, the U.S. has supported the Friends of the Lower Mekong forum, which includes Australia, Japan, South Korea, New Zealand, the European Union, the Asian Development Bank and the World Bank, with the aim of fostering cooperation among the four lower Mekong basin countries.

China's efforts to convert its economic might into increased influence in mainland South East Asia is the most salient feature of Mekong geopolitics. Beijing has long touted the purported benefits of its Lancang dams to those downstream, chiefly through its role in flood control and drought relief. In 2016, for example, Beijing released water at Hanoi's request amid a severe drought. While some interpreted this act as an example of regional cooperation on water management, others saw it as a demonstration that Beijing's could use "coercive water diplomacy" if it were so inclined.[153]

The extent to which China's control of waters in the upper basin represents a coercive lever over lower basin countries is debatable, however. Potentially, Beijing could cut off the entire dry-season flow at the Golden Triangle, where Laos, Myanmar and Thailand meet.[154] But China's dams lack the capacity to impound the total flow of water through consecutive wet and dry seasons.[155] Moreover, the benefits to China of trying to manipulate the Mekong to coerce Vietnam over maritime disputes in the South China Sea, for example, are questionable, since "stopcock diplomacy" would be indiscriminate, inevitably punishing all the other downstream countries as well.[156]

Beijing's construction of massive mainstream dams, carried out without consultation or notification of downstream countries, has complicated its efforts to foster what it calls a "community of common destiny" with its neighbours in mainland South East Asia. Beijing recognises that misgivings in the lower basin about its stewardship of Mekong waters gave the U.S. and its partners an opening in the region.[157] Establishment of China's Lancang-Mekong Cooperation initiative was partly a response to U.S. moves in the Mekong under the rubric of President Barack Obama's "rebalance" toward Asia.

Senior U.S. officials have bluntly criticised China's dam building and infrastructure diplomacy in the Mekong region. Criticism of this ilk was often strident during the Trump administration, with then-Secretary of State Michael Pompeo and others decrying what they called China's "unilateral manipulation" of the river at the expense of downstream countries.[158] Although the Biden administration has not used

standards for infrastructure development". "Chinese dams under U.S. scrutiny in Mekong rivalry", Reuters, 14 December 2020; "The Mekong-U.S. Partnership and the Friends of the Mekong: Proven Partners for the Mekong Region", Office of the Spokesperson, U.S. Department of State, 3 August 2021.

[153] Rabea Brauer and Frederick Kliem, "Coercive Water Diplomacy: Playing Politics with the Mekong", in Gerhard Wahlers (ed.), *International Reports 3: Water, Power, Conflict* (Berlin, 2017), p. 52.

[154] Ian Campbell and Chris Barlow, "Hydropower Development and the Loss of Fisheries in the Mekong River Basin", *Frontiers in Environmental Science*, vol. 8 (October 2020), p. 7.

[155] McPherson, "China's Role in Promoting Transboundary Resource Management in the Greater Mekong Basin", op. cit., p. 10, note 43.

[156] David Hutt, "Water war risk rising on the Mekong", *Asia Times*, 16 October 2019.

[157] Hongzhou Zhang, "China's Water Diplomacy in the Mekong: Drivers and Goals", in Brahma Chellaney and Frederick Kliem (eds.), "The Displaced: Disrupted Trade, Labour and Politics in the Mekong River Basin", Konrad-Adenauer-Stiftung, 2021, p. 43.

[158] "China 'taking control' of vital Mekong River through dam-building spree, Mike Pompeo warns", Agence France-Presse, 2 August 2019; remarks by David R. Stilwell, assistant U.S. secretary of state

the Mekong as a rhetorical cudgel against Beijing, Secretary of State Antony Blinken spoke in August 2021 of a "free and open Mekong", linking the region to a "free and open Indo-Pacific", a strategic paradigm that is largely framed around competition with China.[159]

Controversy over an April 2020 research report by Eyes on Earth, a U.S.-based consultancy, encapsulates the contest between the U.S. and China over which narrative best explains the sources of the Mekong's environmental problems and the causes of tensions between countries in the basin. The study developed a land-surface wetness index based on satellite imagery to compare actual flows, measured by the gauge at Chiang Saen, Thailand, to anticipated flow if there were no Lancang dams.[160] The report and subsequent Stimson Center commentary attributed the 2019 lower Mekong drought primarily to water being held up by China's dams.[161]

The Mekong River Commission and others criticised the report's methodology, arguing that the research did not bear out the conclusions drawn by the Stimson commentary and media reports.[162] The Stimson Center acknowledged errors in the Eyes on Earth data, but it asserted that rerunning the analysis with corrected data confirmed that restriction of water flow by China's dams "exacerbated drought conditions along the course of the Mekong during the 2019 wet season".[163] Chinese officials, academics and media outlets reacted with indignation to the study and associated commentary. The state-run *Global Times* identified what it called an "attack chain formed to peddle the 'Chinese dam threat' over the Mekong River fallacy".[164] In November 2020, Luo Zhaohui, foreign vice minister responsible for Asian affairs,

for East Asian and Pacific affairs, "The Mekong River, Mekong Sovereignty, and the Future of Southeast Asia", Indo-Pacific Conference on Strengthening Transboundary River Governance, 15 October 2020.

[159] "Secretary Blinken's Participation in the East Asia Summit Foreign Ministers' Meeting", Office of the Spokesperson, U.S. Department of State, 4 August 2021.

[160] Alan Basist and Claude Williams, *Monitoring the Quantity of Water Flowing through the Upper Mekong Basin Under Natural (Unimpeded) Conditions* (Bangkok, 2020).

[161] Brian Eyler and Courtney Weatherby, "New evidence: How China turned off the tap on the Mekong River", Stimson Center, 13 April 2020.

[162] Tarek Ketelson et al., "Monitoring the Quantity of Water Flowing Through the Upper Mekong Basin under Natural '(Unimpeded) Conditions'; Rapid Review", Australia-Mekong Partnership for Environmental Resources and Energy Systems, 19 April 2020; "Understanding the Mekong River's Hydrological Conditions: A Brief Commentary Note on the 'Monitoring the Quantity of Water Flowing Through the Upper Mekong Basin Under Natural (Unimpeded) Conditions' Study by Alan Basist and Claude William", MRC Secretariat, 2020; Marko Kallio and Amy Fallon, "Are China's Dams on the Mekong Causing Downstream Drought? The Importance of Scientific Debate", Centre for Social Development Studies, Chulalongkorn University, 28 April 2020.

[163] Brian Eyler et al., "Consultative processes lead to more accurate monitoring of the Mekong from Eyes on Earth and the Stimson Center", Stimson Center, 8 June 2021.

[164] This "attack chain" included "Human Rights Watch, the International Union for Conservation of Nature, the United States Agency for International Development, and The Asia Foundation, [and] media outlets such as *The New York Times* and the BBC". Hu Yuwei, "GT investigates: Who are the mouthpieces of US-led war of public opinion on "Chinese dams' threats" along Mekong River, and what are their typical methods?", *Global Times*, 28 March 2022.

accused external powers of "spread[ing] rumours and exaggerat[ing] China's threats, in a bid to sow discord and undermine cooperation between countries in the region".[165]

## B. *Viewpoints from the Lower Mekong*

The Mekong's portrayal as an arena for Sino-U.S. competition rankles many policymakers and activists in mainland South East Asia, who regard the notion that other countries must choose between China and the U.S. as an affront.[166] The lower Mekong states all confront the dilemma of balancing strong economic ties with China with protecting national interests that may not align with Beijing's unilateral approach to managing the Mekong.[167] But China's proximity and preponderance are brute facts that states seek to turn to their own advantage.

Many policymakers, activists and analysts in the region see the U.S. approach to countering China's influence as needlessly confrontational and lament Washington's tendency to view the region through a China lens.[168] This tendency encourages a perception among analysts and activists that the Mekong-U.S. Partnership is really "political posturing".[169] At the same time, others appreciate U.S. support for programs fostering environmental preservation and transparency, even as they worry about Washington's staying power.[170]

Prior to China's ascent to major-power status over the past 30 years, Thailand was the dominant state in the lower basin due to its geographic position and head start on economic development, at a time when wars were ravaging neighbouring states. Despite early enthusiasm for Chinese hydropower and connectivity, Bangkok has since demonstrated determination to protect its interests, even when these may conflict with Beijing's preferences. In 2020, the Thai cabinet cancelled a longstanding but intermittent Chinese project to clear rapids from the Mekong to allow for navigation by vessels up to 500 tonnes, which Thai civil society groups opposed on environmental grounds. Though a victory for activists, the main factor in the decision was likely Bangkok's concern that navigation improvement schemes, including destruction of rapids and dredging, could result in a loss of territory.[171]

[165] Shi Jiangtao, "China faces uphill struggle to win over Mekong neighbours", *South China Morning Post*, 2 December 2020.
[166] Crisis Group interviews, former diplomat, Bangkok, December 2022; former minister, Phnom Penh, September 2002; environmental activist, Chiang Rai, Thailand, January 2024. This aversion stems in part from the series of Indochina wars, which stretched from the 1940s until the 1990s.
[167] Crisis Group interviews, Ambassador Pou Sothirak, executive director, Cambodian Institute for Cooperation and Peace, Phnom Penh, 14 September 2022; former diplomat, Bangkok, December 2022; academic, Chiang Rai, Thailand, January 2024.
[168] Crisis Group interviews, environment expert, Ho Chi Minh City, August 2022; former minister, Phnom Penh, September 2022; hydropower expert, Chiang Rai, Thailand, January 2024.
[169] Denney et al., "Inclusion in Decision-making around Transboundary Water Governance in the Mekong Sub-region", op. cit., p. 1.
[170] Crisis Group interviews, academic, Phnom Penh, September 2022; activist, Chiang Rai, Thailand, January 2024.
[171] Crisis Group interview, activist, Chiang Rai, January 2024. Busbarat et al., "China and Mekong Regionalism: A Reappraisal of the Formation of Lancang-Mekong Cooperation", op. cit., p. 206. The border is marked by the thalweg, or deepest point in the river, which could shift as result of dams or the impact of navigation improvement schemes. "Thailand openly calls for transboundary impact assessments from Laos' Sanakham dam", *Bangkok Tribune*, 20 January 2021.

Bangkok's unease with China's predominance is reflected in the 2018 revival of the Ayeyarwady-Chao Phraya-Mekong Economic Cooperation Strategy (ACMECS), a grouping of all five South East Asian continental states. Launched in 2003 but soon undermined by Thailand's internal strife, Bangkok revived ACMECS in 2018, announcing a fund for infrastructure development, in part to reduce reliance on China.[172] According to a Thai diplomat, ACMECS intends to help the mainland South East Asian countries achieve a common strategy for dealing with external powers.[173] Western governments have supported ACMECS as an alternative to China's dominance, but Vietnam, perhaps regarding the grouping as a challenge to its influence in Laos and Cambodia, has been hesitant to sign on.[174]

Among Mekong countries, Cambodia and Laos have been the most receptive to Beijing's aid and investment, to the extent that many observers now refer to them as China's client states. In Cambodia, China has backed transport and hydropower infrastructure, as well as the Dara Sakor enclave in Koh Kong province, which includes industrial parks, resorts, a deep-water port and an airport reportedly able to accommodate military aircraft.[175] In May 2023, Cambodia's government approved plans for the $1.7 billion Funan Techo Canal, linking the Mekong south of Phnom Penh to the Gulf of Thailand. The 160km, China-funded canal will allow Cambodia to bypass the Vietnamese ports upon which it currently relies for 30 per cent of its shipping.[176] Beyond the environmental impact on Vietnam's delta, some express concern about the security implications for neighbouring countries of prospective access by Chinese military vessels to the lower Mekong.[177] Beijing's close ties with the ruling Cambodian People's Party have been rewarded with Phnom Penh's support in the South China Sea disputes and access to Ream naval base.[178]

[172] Yukako Ono, "Thailand plans regional infrastructure fund to reduce China dependence", *Nikkei Asia Review*, 4 June 2018; "Thailand spearheads regional fund to cut reliance on big Asian economies", Reuters, 16 June 2018.

[173] Ono, "Thailand plans regional infrastructure fund to reduce China dependence", op. cit.

[174] Crisis Group interview, Western diplomat, Hanoi, December 2022. Susannah Patton, "Thailand's efforts to build Mekong bloc deserve support", *Nikkei Asia Review*, 4 October 2022.

[175] John Bradford, "China's Security Force Posture in Thailand, Laos, and Cambodia", U.S. Institute of Peace, December 2021, p. 16.

[176] Sothearak Sok, "Why is Vietnam worried about Cambodia's Mekong canal project?", *The Diplomat*, 2 January 2024; "Cambodia says it will cut shipping through Vietnam by 70% with new China-funded Mekong canal", Reuters, 7 May 2024.

[177] "Cambodia's canal project would link its coast with the Mekong River", RFA Khmer, 23 January 2024; "US, Vietnam fear China-backed Techo Funan Canal in Cambodia could be used for military purposes", *South China Morning Post*, 11 April 2024.

[178] Crisis Group interviews, Western diplomats, Hanoi, December 2022. Cambodia's objections regarding the issue of the South China Sea prevented agreement on a joint communiqué following the ASEAN Summit in Phnom Penh in 2012. Prime Minister Hun Sen also publicly rejected the 2016 Arbitral Tribunal award in favour of the Philippines. Sovinda Po and Kearrin Sims, "The Myth of Non-interference: Chinese Foreign Policy in Cambodia", *Asian Studies Review* (2021), p. 11. The Ream naval base became a topic of controversy in 2020 after Cambodia razed two U.S.-funded structures at the base and relocated the "Joint Vietnamese Friendship" building from the area in preparation for extensive Chinese-funded renovation of the base. China reportedly operates a ground station for its BeiDou navigation satellite system at Ream though Phnom Penh and Beijing deny it. Ellen Nakashima and Cate Cadell, "China secretly building naval facility in Cambodia, Western officials say", *Washington Post*, 6 June 2022.

Laos is sometimes portrayed as a victim of "debt-trap diplomacy", raising the fear that China can seize Lao infrastructure if it is unable to repay.[179] Over the past five years, the Lao economy has struggled under massive public debt, estimated at 122 per cent of GDP in 2023. About half of that debt belongs to China, which has provided funds under the Belt and Road Initiative for large infrastructure projects, notably the high-speed railway linking Vientiane (and eventually Thailand) to Yunnan, which cost $6 billion. China also backs large hydropower development in Laos. In September 2020, state-owned Électricité du Laos partnered with China Southern Power Grid Company to create the Électricité du Laos Transmission Company. China Southern, wholly owned by China's government, has a controlling share of the firm, which operates the grid under a 25-year concession.[180]

China's outsized influence in Laos and Cambodia is a concern for Vietnam, which has traditionally counted these countries within its sphere of influence.[181] Vientiane's decision to build the Xayaburi dam despite Hanoi's objections was particularly significant, marking the first time since 1975 that Laos had publicly disagreed with Vietnam.[182] In parallel, Hanoi is worried about encirclement, due to China's assertiveness in the South China Sea but also to its newfound access to Cambodia's Ream naval base and, potentially, the Funan Techo Canal. According to a Vietnamese scholar, "Vietnam is being squeezed in a vice between the [South China] Sea and the Mekong".[183] Hanoi has accordingly pursued seemingly contradictory policies: while mobilising to deal with the threats to the delta posed by upstream dams, it continues to invest in and import Lao hydroelectricity that exacerbate the problem because it is unwilling to cede its influence in Laos.

During its chairmanship in 2020 of the Association of South East Asian Nations (ASEAN), the region's most important inter-governmental forum, Vietnam's efforts to place the Mekong on the agenda gained little traction, and subsequent chairs have sidestepped the issue. With the four lower Mekong states lacking a common position, other ASEAN member states are reluctant to table Mekong issues that could antagonise Beijing, reflecting a divide between mainland and maritime members.[184] The grouping's own Mekong initiative, ASEAN-Mekong Basin Development Cooperation, intended to reduce development disparities between member states, has been moribund for at least a decade.

[179] Some argue that China is unlikely to allow Laos to default, as it would damage its own reputation as a lender. China already provided Laos with near-term debt relief from 2020-2022, with deferred payments amounting to 8 per cent of GDP. Jittipat Poonkham, "Power Politics and the Institutional Architecture in the Mekong Subregion: Beyond the Geopolitical Trap?", *International Studies Centre Journal*, no. 1 (June 2022), p. 3; Toshiro Nishizawa, "Claims of default in Laos are bankrupt", East Asia Forum, 25 August 2022; Nyshka Chandran, "Laos is spiraling toward a debt crisis as China looms large", CNBC, 8 November 2023.

[180] Mariza Cooray, "What price will Laos need to pay to be saved, and will China pay it?", The Interpreter (blog), 2 May 2023.

[181] Crisis Group interviews, Western diplomats, Hanoi, December 2022.

[182] Crisis Group interview, Ian Baird, scholar, Chiang Mai, Thailand, January 2024.

[183] Crisis Group interview, Ho Chi Minh City, Vietnam, August 2022.

[184] Crisis Group interview, former diplomat, Bangkok, December 2022; Hoang Thi Ha and Farah Nadine Seth, "The Mekong River Ecosystem in Crisis: ASEAN Cannot be a Bystander", Fulcrum, 1 June 2021.

## VI. Leeway for Change

Large-scale infrastructure such as hydropower dams often benefits elites with decision-making power, providing them with prestige, sizeable investment returns and opportunities for graft. Information that is at odds with the interests of policymakers tends to get short shrift. As a hydropower expert put it: "These officials have heard it all before. They don't care about the flood pulse or sediment".[185] Yet the potential impact of looming environmental catastrophe, exacerbated by climate change, is likely to be so far-reaching that the costs will not remain confined to the poor and marginalised. To avoid a catastrophe in the Mekong basin, policymakers will have to reconsider the fundaments of their decision-making on infrastructure and the role of transboundary governance.

One bright spot in the Mekong quandary is that there is already an abundance of policy and technical guidance on how to mitigate the most damaging effects of large-scale infrastructure and increase resilience to climate change. To some recommendations – such as data sharing and energy transition, for example – policymakers have already proven receptive. Others, such as participatory policymaking and regional policy coordination, are more ambitious for Mekong governments, at least for the moment. Still, there are steps in these directions that can be taken, notwithstanding the impediments in the way of changing policy across the region.

### A. *Energy Transition*

Lower Mekong states have made public commitments to reduce greenhouse gas emissions and make the transition to carbon-neutral energy (a balance between carbon dioxide emissions and absorption) by 2050.[186] Studies show that it is economically and technically feasible for the region to meet emissions and energy targets while reducing or even halting planned large hydropower development.[187] With adequate investments, the lower Mekong basin countries can drastically reduce or eliminate emissions and meet rising energy demand by 2050 through a combination of solar, wind, biomass and off-river pumped hydropower generation, which cycles water between higher and lower reservoirs.[188] Large-scale hydropower plans should be shelved

[185] Crisis Group telephone interview, hydropower expert, July 2022. He added that it would be more effective to find alternative sources of rent to replace those from large hydropower projects than to seek accountability from officials.

[186] At the UN's annual climate conference in 2021, Thailand set a national target of carbon neutrality by 2050 and net-zero emissions by 2065, while Vietnam committed to net-zero emissions by 2050. Cambodia also pledged carbon neutrality by 2050, and Laos pledged to unconditionally reduce its greenhouse gas emissions by 60 per cent compared to business as usual by 2030 and achieve net zero by 2050. Plans for meeting these targets are codified in national power development and energy strategies.

[187] Kais Siala et al., "Solar Energy and Regional Coordination as a Feasible Alternative to Large Hydropower in Southeast Asia", *Nature Communications* (2021), p. 6.

[188] Intelligent Energy Systems and Mekong Economics, "Alternatives for Power Generation in the Greater Mekong Subregion, Volume 1: Power Sector Vision for the Greater Mekong Subregion", 5 April 2016; Rathana Lorm and Bundit Limmeechokchai, "Energy System Transformation to Net Zero CO2 Emissions 2050 in Selected GMS Countries", *International Energy Journal*, vol. 23 (September 2023); Hon Chung Lau, "Decarbonization of ASEAN's Power Sector: A Holistic Approach",

in favour of these energy sources, which do not have the same environmental and social costs.

Achieving this transformation will be challenging. While each state faces its own impediments, some imperatives are common to all.[189] It requires billions of dollars in investment, regulatory reform, innovation in energy storage, improvements to national grids and enhanced regional grid connectivity. Above all, it requires policymakers to overcome entrenched interests in business as usual.[190]

The question of how to finance energy transition in developing economies exemplifies the need for policymakers to act boldly. To achieve net zero by 2050, annual capital spending in developing economies globally (excluding China) must increase sevenfold to $1 trillion by the end of the 2020s. Though a huge sum, it is "small compared to the costs of inaction and the size of the global economy ($160 trillion in 2022)", according to the World Bank.[191] Wealthy countries and international financial institutions need to develop sustainable finance frameworks to channel funding to alternative energy sources and emerging and developing economies in a more equitable way.[192] As a biodiversity hotspot, the lower Mekong basin should be a priority region for such projects.

## B. *Regional Coordination*

Serious efforts to halt and reverse the ecological harm of large hydropower and other infrastructure require basin-wide planning and action. Deeper coordination between the Mekong countries is made harder by huge disparities in power and wealth, as well as the differing effects of infrastructure projects upon them. Rhetorically, at least, lower basin governments recognise the need to work more closely together. At the fourth Mekong River Commission summit in 2023, members states pledged "to move

*Energy Reports*, vol. 9 (2023); Thang Nam Do et al., "Harnessing Solar and Wind for Sustainable Cross-border Electricity Trade in the Greater Mekong Subregion", *Frontiers in Environmental Science* (2023).

[189] Vietnam's recent experience with solar and wind generation is instructive. Eager to expand renewable energy output, Vietnam went from 997 gigawatt hours in 2018 to an 37,865 GWh in 2022 after introducing attractive feed-in tariffs. But the boom was followed by a bust. A short window for approval of generous feed-in tariffs and uncertainty about the future tariff regime resulted in rushed project proposals. Power Purchase Agreements with state-owned Electricity Vietnam, the country's sole off-taker, are risky for investors, offering no purchase guarantee. Problems with grid capacity and system stability also led to curtailment, with projects forced to operate at 30 to 60 per cent capacity, causing financial losses for many. According to an energy expert, "it was too much, too fast". Crisis Group interview, Ho Chi Minh City, Vietnam, August 2022. Thang Nam Do et al., "Vietnam's Solar and Wind Power Success: Policy Implications for the other ASEAN Countries", *Energy for Sustainable Development*, no. 65, (2021), p. 1; Le Hong Hiep, "The Unexpected Twist in Vietnam's Renewable Energy Saga", Fulcrum, 4 January 2024.

[190] Carl Middleton and Tarek Ketelson, "Pathways to a Sustainable and Just Transformation of the Mekong Region's Electricity Sector", in Carl Middleton and Tarek Ketelson (eds.), *Pathways to a Sustainable and Just Transformation of the Mekong Region's Electricity Sector* (Bangkok, 2022), p. 7.

[191] "Scaling Up to Phase Down: Financing Energy Transitions in the Power Sector", World Bank, 2023, p. iv.

[192] "Financing Clean Energy Investment in Emerging and Developing Economies", International Energy Agency, 2021, p. 16; International Renewable Energy Agency, *World Energy Transitions Outlook 2023: 1.5°C Pathway*, 2023, p. 26.

beyond water-resources planning to encompass operational management, including … transboundary coordination".[193] Yet for now the MRC's basin development strategies remain aspirational and development continues piecemeal, following national plans and investors' interests.[194]

Technically, it is possible to coordinate operation of dams in the lower Mekong basin to mimic in some degree the river's natural flow, without sacrificing overall power generation.[195] Decommissioning the most environmentally harmful and least efficient dams would help in this endeavour. But the political and regulatory impediments to such coordinated dam re-operation make this option unrealistic at present.[196] There has, however, been movement toward realising a long-held ambition to build a Mekong regional power grid. A regional high-voltage direct current grid would facilitate power trade and increase energy efficiency, connecting load centres to new renewable electricity production sites, thereby reducing energy storage needs.[197] The Lao PDR-Thailand-Malaysia-Singapore Power Integration Project, the first multilateral power trade arrangement in South East Asia, is a step in this direction.[198] Lower Mekong basin states should seek to harmonise technical standards and operational procedures to facilitate the regional energy trade, in accord with the Intergovernmental Agreement on Regional Power Trade signed by Mekong basin countries in 2004 under the auspices of the Greater Mekong Subregion Initiative.[199]

### C. *Research and Data Sharing*

Data collection and dissemination are crucial for better management of hydropower in the Mekong basin and planning to mitigate its adverse impact, particularly flooding. Data sharing improves water management by helping avoid unanticipated water shocks, curbing investment in inappropriate infrastructure and building trust between water users.[200]

While sharing of water data between Mekong River Commission members states has been standard institutional practice since 2001, China has traditionally been re-

[193] Vientiane Declaration, "Innovation and Cooperation for a Water Secure and Sustainable Mekong", 5 April 2023.
[194] "Annual Report 2022: Key Achievements and Impact Stories", Mekong River Commission Secretariat, 2023, p. xvii.
[195] Stefano Galelli et al., "Opportunities to Curb Hydrological Alterations via Dam Re-operation in the Mekong", *Nature Sustainability*, vol. 5 (December 2022).
[196] Crisis Group interview, European Union diplomat, Hanoi, December 2022.
[197] Do et al., "Harnessing Solar", op. cit., p. 4; Galelli et al., "Opportunities to Curb Hydrological Alterations via Dam Re-operation in the Mekong", op. cit., p. 1064.
[198] Under this project, Singapore has imported Lao electricity via Thailand and Malaysia since June 2022. Sharon Seah and Mirza Sadaqat Huda, "Connectivity and Resilience in the ASEAN Power Grid 2.0 under Laos' ASEAN Chairmanship", Fulcrum, 26 January 2024.
[199] Thang Nam Do and Paul J. Burke, "Is ASEAN Ready to Move to Multilateral Cross-border Electricity Trade?", *Asia Pacific Viewpoint* (2022), p. 8. The Intergovernmental Agreement on power trade established the Regional Power Trade Coordinating Committee. The committee has two working groups, on regulatory issues and performance standards and grid code.
[200] Carl Middleton et al., *Strengthening Water Diplomacy through Water Data Sharing and Inclusive Evidence-based Transboundary Governance*, Center for Social Development Studies, Chulalongkorn University, and Cambodia Development Research Institute, August 2021, p. 22.

luctant to divulge information about its dam operations, treating them as a matter of national security.[201] Over the past fifteen years, however, Beijing has progressively expanded data sharing with the Commission and its member states. In 2002, China and the MRC signed an agreement on technical cooperation in which China agreed to share water level and rainfall data from Manwan and Jinghong dams on the upper Mekong during the flood season. In October 2020, amid extreme drought and growing pressure for greater transparency, Beijing agreed to expand data sharing from these two stations to twice daily year-round.[202] It is an improvement, but a long way from the "effective near real-time sharing of storage levels and hydropower operations data", including from tributaries, endorsed by senior Mekong country officials in Beijing in September 2023.[203]

Data sharing needs to go much further. Above all, it should enable the communities most affected by environmental degradation in the Mekong basin, including women and Indigenous Peoples, to share information from their localities and for this data to strengthen early warning systems, guide policymaking and encourage accountability.[204] Affected communities should also be able to gain access to environmental data through user-friendly portals.[205]

Research, data collection and monitoring are also fundamental to addressing the problems arising from excessive sand mining in the region, especially in the delta. The first step is to measure how much sand is introduced into the basin and how much is extracted, using remote sensing and sonar technology.[206] National Mekong Committees should track sand-mining statistics and the Mekong River Commission Secretariat should incorporate sand monitoring in its Core Monitoring Function Division.[207] This information can inform sand budgets detailing the balance of sand

[201] The MRC approved Procedures for Data and Information Exchange and Sharing in 2001 to facilitate exchange of data between member states, managed by the MRC Secretariat, on water resources, agriculture, navigation, flood management and ecology, among others. "Procedures for Data and Information Exchange and Sharing", Mekong River Commission, 1 November 2001. The MRC overhauled its Core River Monitoring Network in 2022 to address shortcomings in data sharing, including deficient technical capabilities and ineffective integration among monitoring stations. "MRC Council acts to strengthen regional cooperation, protection of Mekong", *Laotian Times*, 28 November 2022. On China, see Magee, "The Dragon Upstream", op. cit., p. 190.

[202] "Mekong River levels 'slightly' rise amid conflicting regional data", Reuters, 22 February 2021.

[203] "Technical Report – Phase 1 of the Joint Study on the Changing Patterns of Hydrological Conditions of the Lancang-Mekong River Basin and Adaptation Strategies", Lancang-Mekong Water Resources Cooperation Center and Mekong River Commission, 2023, p. 79; "Mekong-Lancang countries reach agreement on dam operation data sharing", Vietnam News Agency, 12 September 2023.

[204] "Democratising data for more inclusive governance to strengthen conservation in the Mekong Basin", IUCN, 20 April 2023.

[205] P. Chung et al., *Mapping Knowledge and Gaps in Data, Transparency and Capacity Building Practices in the Mekong Region*, Open Development Initiative, World Wide Fund for Nature Greater Mekong Hub (2022), p. 2; "Democratising data for more inclusive governance to strengthen conservation in the Mekong Basin", op. cit.

[206] Rachel Yu San Lau and Edward Park, "Riverbed sand mining threatens the future stability of the Vietnamese Mekong Delta", Global Water Forum, 23 November 2023; Edward Park, "Sand Mining in the Mekong Delta: Extent and Compounded Impacts", *Science of the Total Environment*, no. 924 (2024), p. 14.

[207] "Mekong-U.S. Partnership Track 1.5 Policy Dialogue on Transboundary Water Governance

entering and leaving a given stretch of river and identification of areas where sand can be extracted without exceeding replenishment rates.[208] National and local governments will have to promulgate and strictly enforce regulations on where and how much sand may be mined.[209]

### D. *Participatory Governance*

The Mekong region's governments offer few meaningful avenues for public input into decision-making about development objectives, large-scale infrastructure and energy planning.[210] Too often, public engagement is absent or minimal, undertaken as a matter of form to ratify development plans already in train and backed by powerful state and commercial interests.[211] Women often face additional obstacles to their participation, among other because of domestic responsibilities and more limited formal education.[212] Ensuring that the preferences and concerns of local communities and marginalised groups, who are most directly touched by large-scale infrastructure projects, are heard and considered in their planning and roll-out would allow for decision-making that better factors in public well-being.

Participatory governance of development is an inherently political matter, whereas existing mechanisms across the Mekong tend to treat development objectives and infrastructure plans as technical matters.[213] More entry points for public participation are needed in national energy planning and the Mekong River Commission's consultation procedures. The Commission's membership should be broadened to include civil society representatives. States and investors should observe the principles of free, prior and informed consent for local communities affected by infrastructure development, which should also help to forestall potential legal challenges and reputational costs. They could also subject development proposals to criticism by communities and compare them with alternatives.[214] Beyond access to opportunities to influence decision-making, people and communities must be able to express their views without fear of retribution.

Summary Report", Stimson Center and International Union for Conservation of Nature (September 2023), p. 5.

[208] Lau and Park, "Riverbed sand mining", op. cit.

[209] Park, "Sand Mining in the Mekong Delta: Extent and Compounded Impacts", op. cit., p. 16.

[210] Crisis Group interviews, activists, Chiang Rai and Chiang Khong, Thailand, January 2024. Grumbine, op. cit., p. 9.

[211] Chattranond, op. cit., p. 150.

[212] Delfau and Yeophantong, op. cit., pp. 19, 45.

[213] Hirsch, "IWRM as a Participatory Governance", op. cit., p. 168.

[214] John Dore and Louis Lebel, "Deliberation and Scale in Mekong Region Water Governance", *Environmental Management*, vol. 46 (2010), p. 60.

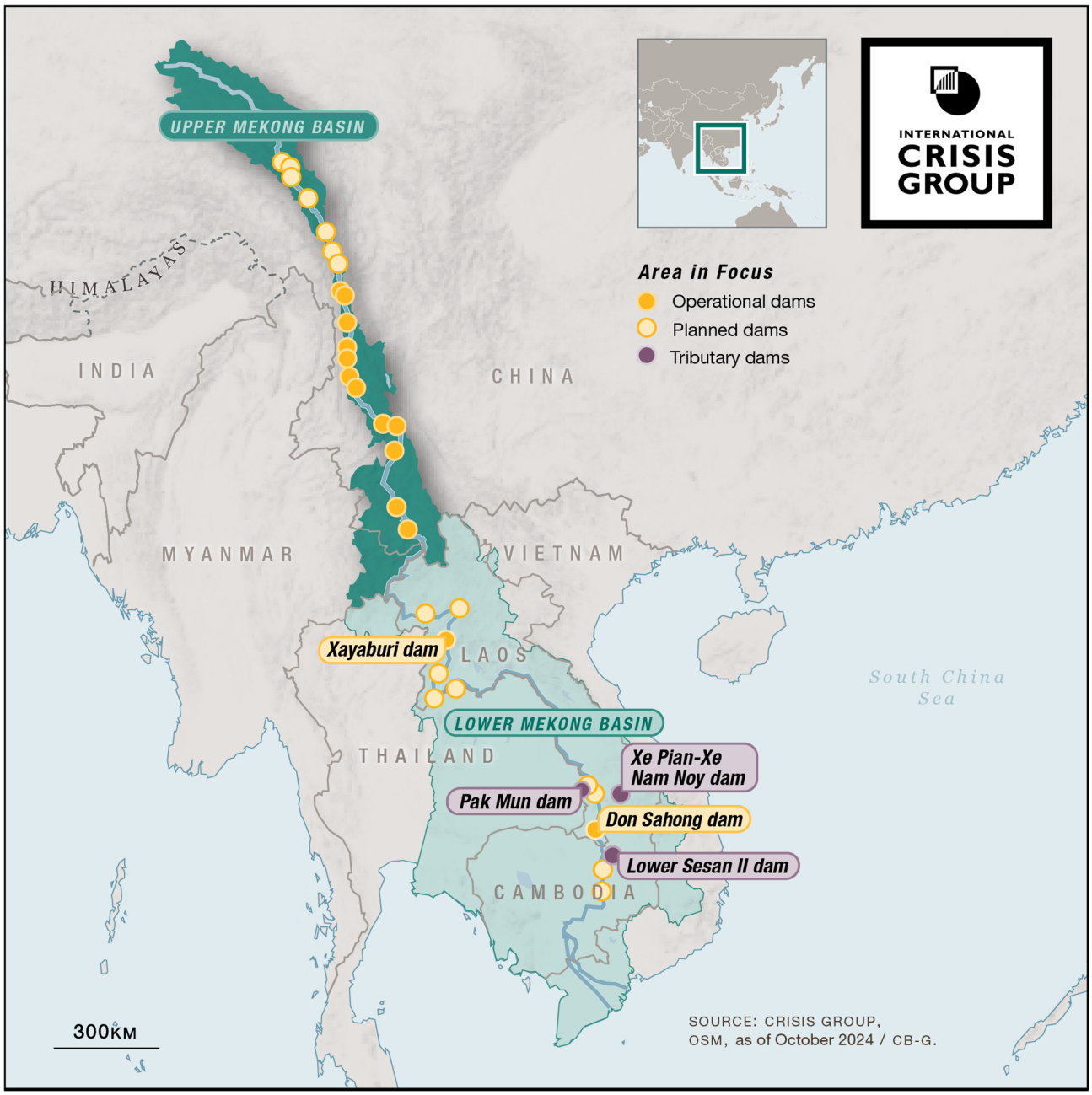
## VII. Conclusion

The number of interests at play, disparities in power and wealth, and the general lack of influence over decision-makers for those hit hardest by changing conditions in the lower Mekong basin all point to a deepening environmental crisis and increasing insecurity in food, water and livelihoods. The absence of more friction between states, despite their divergent interests, is attributable in part to how the costs and benefits of infrastructure development are distributed. For now, rural communities who depend on the natural flow of the Mekong and its tributaries pay the costs, while investors and state agencies reap most of the rewards. This imbalance is unlikely to hold indefinitely. The scope of prospective ecological decline is so great that it threatens not only the poor and marginalised, who have traditionally borne the brunt of the harm from large-scale infrastructure building, but also potentially the elite that most benefit from these projects. More broadly, it threatens the sustainability of economic growth and the lives of dozens of millions of people.

The nexus of problems, cumulative and mutually reinforcing, are tremendously hard to resolve under current political conditions. But there is still leeway for improvement. Technological advances in energy generation and storage hold out some hope. Greater coordination between states, increased data sharing and more inclusive policy processes are within the realm of possibility. But, as so often with environmental concerns, a change in mindsets among policymakers who are more preoccupied with near-term goals than the slowly emerging consequences of over-exploiting the Mekong appears to be a necessary precondition for far-reaching reform. Hard experience can help change minds, but usually too late and at too great a cost. Anticipation and action to avert the worst is the wiser course.

**Bangkok/Brussels, 7 October 2024**

## Appendix A: Map of Lancang/Mekong Mainstream and Selected Tributary Hydropower Dams



## Appendix B: Mekong Hydrology and Geography

The Mekong River rises in the Tibetan Plateau, dropping 4,500m in elevation over 2,400km through narrow gorges before leaving China to form a 100km border between Myanmar and Laos.[215] In China, the river is known as Lancang Jiang. From the Golden Triangle, where Myanmar, Laos and Thailand meet, the river traces the Thai-Lao border for 97km before bearing east, coursing exclusively through Laos. The river again forms the Thai-Lao border roughly 150km west of Vientiane, then bounds the Khorat Plateau of north-eastern Thailand before drifting into southern Laos. The river spills over rapids and cataracts of the Khone Falls into northern Cambodia, where the river begins to widen as highlands give way to the Mekong floodplain that is annually inundated in the wet season. The Mekong Delta begins south of Phnom Penh, where the river flows south east, forming two branches as it crosses into southern Vietnam, the Bassac to the west and the Mekong to the east.[216] From the Delta the river courses through multiple channels and empties into the South China Sea, discharging 475 cubic km annually.[217]

The river's hydrology derives principally from runoff from two distinct regions, namely, snowmelt from the Tibetan Plateau and tropical tributaries mostly from the river's left bank in Laos.[218] The upper basin contributes about 18 per cent of the river's total discharge, with 2 per cent coming from Myanmar tributaries.[219] Contributions from these two zones vary seasonally. In the upper part of the lower Mekong, from the Golden Triangle to Vientiane, the Yunnan component provides most of the flow, as much as 75 per cent during the dry season in April and May and over 50 per cent during peak flow months July-September. In the lower stretches, at Kratie in Cambodia, for example, China's contribution is roughly 15-20 per cent during the flood season, but as much as 40 per cent in the dry season.[220]

[215] Avijit Gupta, "Geology and Landforms of the Mekong Basin", in Ian Campbell (ed.), *The Mekong: Biophysical Environment of an International River Basin* (New York, 2009), p. 42.
[216] Ibid., pp. 42-47.
[217] "The Flow of the Mekong", Mekong River Commission, November 2009, p. 2.
[218] Peter T. Adamson, Ian D. Rutherfurd, Murray C. Peel and Iwona A. Conlan, "The Hydrology of the Mekong River", in Campbell, *The Mekong*, op. cit., p. 54.
[219] Ibid., p. 57.
[220] Ibid., pp. 56-57.

## Appendix C: About the International Crisis Group

The International Crisis Group (Crisis Group) is an independent, non-profit, non-governmental organisation, with some 120 staff members on five continents, working through field-based analysis and high-level advocacy to prevent and resolve deadly conflict.

Crisis Group's approach is grounded in field research. Teams of political analysts are located within or close by countries or regions at risk of outbreak, escalation or recurrence of violent conflict. Based on information and assessments from the field, it produces analytical reports containing practical recommendations targeted at key international, regional and national decision-takers. Crisis Group also publishes *CrisisWatch*, a monthly early-warning bulletin, providing a succinct regular update on the state of play in up to 80 situations of conflict or potential conflict around the world.

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Comfort Ero was appointed Crisis Group's President & CEO in December 2021. She first joined Crisis Group as West Africa Project Director in 2001 and later rose to become Africa Program Director in 2011 and then Interim Vice President. In between her two tenures at Crisis Group, she worked for the International Centre for Transitional Justice and the Special Representative of the UN Secretary-General in Liberia.

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**October 2024**

## Appendix D: Reports and Briefings on Asia since 2021

### Special Reports and Briefings

*Ten Challenges for the UN in 2021-2022*, Special Briefing N°6, 13 September 2021.

*7 Priorities for the G7: Managing the Global Fallout of Russia's War on Ukraine*, Special Briefing N°7, 22 June 2022.

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